



THE UNIVERSITY  
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# A primer on importance sampling and particle filters

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# Outline

## State-space models and Bayesian filtering

### Importance sampling: basics and advanced methods

- Monte Carlo basics

- Importance sampling (IS) basics

- Multiple importance sampling (MIS)

- Adaptive importance sampling (AIS)

### Particle filtering

- From IS to sequential IS

- Mini-project

- In search of better particle filters

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### Particle filtering from the MIS and AIS perspectives

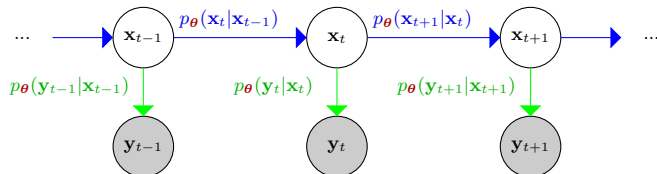
### Advanced particle filtering

- PFs in high-dimension spaces

- The choice of  $N$  in PFs

# 1. Modeling: state-space models (SSM)

- Time-series data are collected,  $\mathbf{y}_t \in \mathbb{R}^{N_y}$ ,  $t = 1, \dots, T$ :
- A SSM models a sequence of hidden states  $\mathbf{x}_t \in \mathbb{R}^{N_x}$ ,  $t = 1, \dots, T$ .



- Probabilistic notation of a (simple) Markovian SSM:
  - state model  $\rightarrow p_{\theta}(\mathbf{x}_t | \mathbf{x}_{t-1}) = p(\mathbf{x}_t | \mathbf{x}_{t-1}, \theta)$
  - observation model  $\rightarrow p_{\theta}(\mathbf{y}_t | \mathbf{x}_t) = p(\mathbf{y}_t | \mathbf{x}_t, \theta)$
  - prior on initial state  $\rightarrow p_{\theta}(\mathbf{x}_0) = p(\mathbf{x}_0 | \theta)$

## 2. Estimation/inference problems

- We sequentially observe data  $\mathbf{y}_t$  related to the hidden state  $\mathbf{x}_t$ .
  - At time  $t$ , we have accumulated  $t$  observations,  $\mathbf{y}_{1:t} \equiv \{\mathbf{y}_1, \dots, \mathbf{y}_t\}$ .
- Wish list:
  - prediction of future **observations** and estimation of **states** (with uncertainty quantification)
    - \* **Filtering**:  $p_{\theta}(\mathbf{x}_t | \mathbf{y}_{1:t})$  and joint  $p_{\theta}(\mathbf{x}_{1:t} | \mathbf{y}_{1:t})$
    - \* State prediction:  $p_{\theta}(\mathbf{x}_{t+\tau} | \mathbf{y}_{1:t})$ ,  $\tau \geq 1$
    - \* Observation prediction:  $p_{\theta}(\mathbf{y}_{t+\tau} | \mathbf{y}_{1:t})$ ,  $\tau \geq 1$
    - \* **Smoothing**:  $p_{\theta}(\mathbf{x}_{t-\tau} | \mathbf{y}_{1:t})$ ,  $\tau \geq 1$
  - estimation of **model parameters** (with interpretability)
- Bayesian/probabilistic inference:
  - we compute or approximate pdfs of unknowns when possible (instead of point-wise estimates)



# Bayesian filtering

- Bayesian rule for the joint:

$$p(\mathbf{x}_{1:T}|\mathbf{y}_{1:T}) = \frac{p(\mathbf{y}_{1:T}|\mathbf{x}_{1:T})p(\mathbf{x}_{1:T})}{p(\mathbf{y}_{1:T})}$$

- Filtering distribution as a marginal:

$$p(\mathbf{x}_T|\mathbf{y}_{1:T}) = \int p(\mathbf{x}_{1:T}|\mathbf{y}_{1:T})d\mathbf{x}_1d\mathbf{x}_2\dots d\mathbf{x}_{T-1}$$

- Problems:

- **Dimension:**  $\mathbf{x}_{1:T} \in \mathbb{R}^{T \cdot d_x}$
- When we receive  $\mathbf{y}_t$ , we don't want to **reprocess**  $\mathbf{y}_{1:t-1}$

Goal: **efficient and sequential** Bayesian inference

# Sequential optimal filtering

- **Filtering Problem:**
  - Distribution of  $\mathbf{x}_t$  given all the obs. up to time  $t$ ,  $p(\mathbf{x}_t|\mathbf{y}_{1:t})$
  - Recursively from  $p(\mathbf{x}_{t-1}|\mathbf{y}_{1:t-1})$  updating with the new  $\mathbf{y}_t$

- Optimal filtering:

1. **Prediction** step:

$$p(\mathbf{x}_t|\mathbf{y}_{1:t-1}) = \int p(\mathbf{x}_t|\mathbf{x}_{t-1})p(\mathbf{x}_{t-1}|\mathbf{y}_{1:t-1})d\mathbf{x}_{t-1}$$

2. **Update** step:

$$p(\mathbf{x}_t|\mathbf{y}_{1:t}) = \frac{p(\mathbf{y}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{y}_{1:t-1})}{p(\mathbf{y}_t|\mathbf{y}_{1:t-1})}$$

- Interest in integrals of the form:  $I(f) = \int f(\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{y}_{1:t})d\mathbf{x}_t$ 
  - e.g., the mean,  $I(f) = \int \mathbf{x}_t p(\mathbf{x}_t|\mathbf{y}_{1:t})d\mathbf{x}_t$
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# Linear-Gaussian state-space model

- Kalman filter and RTS smoother are the close-form solutions for the linear-Gaussian SSM:

- Hidden state  $\rightarrow p(\mathbf{x}_t|\mathbf{x}_{t-1}) \equiv \mathcal{N}(\mathbf{x}_t; \mathbf{A}_t\mathbf{x}_{t-1}, \mathbf{Q}_t)$
- Observations  $\rightarrow p(\mathbf{y}_t|\mathbf{x}_t) \equiv \mathcal{N}(\mathbf{y}_t; \mathbf{H}_t\mathbf{x}_t, \mathbf{R}_t)$

## Kalman filter

- Initialize:  $\mathbf{m}_0, \mathbf{P}_0$
- For  $t = 1, \dots, T$

Predict stage:

$$\mathbf{x}_t^- = \mathbf{A}_t \mathbf{m}_{t-1}$$

$$\mathbf{P}_t^- = \mathbf{A}_t \mathbf{P}_{t-1} \mathbf{A}_t^\top + \mathbf{Q}_t$$

Update stage:

$$\mathbf{z}_t = \mathbf{y}_t - \mathbf{H}_t \mathbf{x}_t^-$$

$$\mathbf{S}_t = \mathbf{H}_t \mathbf{P}_t^- \mathbf{H}_t^\top + \mathbf{R}_t$$

$$\mathbf{K}_t = \mathbf{P}_t^- \mathbf{H}_t^\top \mathbf{S}_t^{-1}$$

$$\mathbf{m}_t = \mathbf{x}_t^- + \mathbf{K}_t \mathbf{z}_t$$

$$\mathbf{P}_t = \mathbf{P}_t^- - \mathbf{K}_t \mathbf{S}_t \mathbf{K}_t^\top$$

## RTS smoother

- For  $t = T, \dots, 1$

Smoothing stage:

$$\mathbf{x}_{t+1}^- = \mathbf{A}_t \mathbf{m}_t$$

$$\mathbf{P}_{t+1}^- = \mathbf{A}_t \mathbf{P}_t \mathbf{A}_t^\top + \mathbf{Q}_t$$

$$\mathbf{G}_t = \mathbf{P}_t \mathbf{A}_t^\top (\mathbf{P}_{t+1}^-)^{-1}$$

$$\mathbf{m}_t^s = \mathbf{m}_t + \mathbf{G}_t (\mathbf{m}_{t+1}^s - \mathbf{x}_{t+1}^-)$$

$$\mathbf{P}_t^s = \mathbf{P}_t + \mathbf{G}_t (\mathbf{P}_{t+1}^s - \mathbf{P}_{t+1}^-) \mathbf{G}_t^\top$$

✓ Filtering distribution:  $p(\mathbf{x}_t|\mathbf{y}_{1:t}) = \mathcal{N}(\mathbf{x}_t; \mathbf{m}_t, \mathbf{P}_t)$

✓ Smoothing distribution:  $p(\mathbf{x}_t|\mathbf{y}_{1:T}) = \mathcal{N}(\mathbf{x}_t; \mathbf{m}_t^s, \mathbf{P}_t^s)$



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# Monte Carlo basics

- Goal:<sup>1</sup> approximate the integral

$$I(f) \equiv \mathbb{E}_{\pi(\mathbf{x})}[f(\mathbf{x})] = \int f(\mathbf{x})\pi(\mathbf{x})d\mathbf{x}$$

- If we can sample  $\mathbf{x}^{(n)} \sim \pi(\mathbf{x})$ ,  $n = 1, \dots, N$ , then the target can be approximated as

$$\pi(\mathbf{x}) \approx \pi^N(\mathbf{x}) = \frac{1}{N} \sum_{n=1}^N \delta_{\mathbf{x}^{(n)}}(\mathbf{x})$$

and the moment  $I(f)$  can be approximated as

$$I(f) \approx \bar{I}_N(f) = \frac{1}{N} \sum_{n=1}^N f(\mathbf{x}^{(n)})$$

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<sup>1</sup>C. P. Robert, G. Casella, and G. Casella. *Monte Carlo statistical methods*. Vol. 2. Springer, 1999.

- This is called raw/crude/direct/standard Monte Carlo.

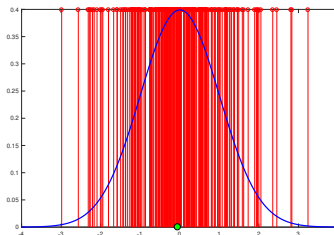
## Monte Carlo basics

- We can approximate any integral involving  $\pi(\mathbf{x})$ : **integral + dirac = sum!**

$$\begin{aligned} \mathbb{E}_{\pi(\mathbf{x})}[f(\mathbf{x})] &= \int f(\mathbf{x})\pi(\mathbf{x})d\mathbf{x} \approx \int f(\mathbf{x})\pi^N(\mathbf{x})d\mathbf{x} \approx \int f(\mathbf{x}) \left( \frac{1}{N} \sum_{n=1}^N \delta_{\mathbf{x}^{(n)}}(\mathbf{x}) \right) d\mathbf{x} \\ &\approx \frac{1}{N} \sum_{n=1}^N \int f(\mathbf{x}) \delta_{\mathbf{x}^{(n)}}(\mathbf{x}) d\mathbf{x} \\ &= \frac{1}{N} \sum_{n=1}^N f(\mathbf{x}^{(n)}) \end{aligned}$$

- e.g. the mean of the distribution ( $h(\mathbf{x}) = \mathbf{x}$ ) approximated by  $N = 200$ .

$$\hat{\mathbf{x}} = \mathbb{E}_{\pi(\mathbf{x})}[\mathbf{x}] = \int \mathbf{x}\pi(\mathbf{x})d\mathbf{x} \approx \frac{1}{N} \sum_{n=1}^N \mathbf{x}^{(n)} = -0.0561$$



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# Importance sampling (IS) basics

- Unfortunately, we only know how to draw samples from very few distributions.
  - In the general case, we cannot use the raw Monte Carlo.
  - Importance sampling (IS)** is a Monte Carlo method that allows to approximate integrals over complicated distributions.<sup>2</sup>
- Same problem:

$$I(f) \equiv \mathbb{E}_{\pi(\mathbf{x})}[f(\mathbf{x})] = \int f(\mathbf{x})\pi(\mathbf{x})d\mathbf{x} = \int f(\mathbf{x})\frac{\pi(\mathbf{x})}{q(\mathbf{x})}q(\mathbf{x})d\mathbf{x}$$

where  $q(\mathbf{x})$  is the proposal density where the samples are now drawn

$$\text{Standard MC estimator: } \bar{I}_N(f) = \frac{1}{N} \sum_{n=1}^N f(\mathbf{x}^{(n)}), \quad \mathbf{x}^{(n)} \sim \pi(\mathbf{x})$$

$$\text{Basic IS estimator: } \hat{I}_N(f) = \frac{1}{N} \sum_{n=1}^N f(\mathbf{x}^{(n)}) \frac{\pi(\mathbf{x}^{(n)})}{q(\mathbf{x}^{(n)})}, \quad \mathbf{x}^{(n)} \sim q(\mathbf{x})$$

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<sup>2</sup>V. Elvira and L. Martino. "Advances in importance sampling". In: *arXiv preprint arXiv:2102.05407* (2021).

# Importance sampling basics

- Basic IS estimator:

$$\hat{I}_N(f) = \frac{1}{N} \sum_{n=1}^N f(\mathbf{x}^{(n)}) \frac{\pi(\mathbf{x}^{(n)})}{q(\mathbf{x}^{(n)})}, \quad \mathbf{x}^{(n)} \sim q(\mathbf{x})$$

- $W^{(n)} = \frac{\pi(\mathbf{x}^{(n)})}{q(\mathbf{x}^{(n)})}$ ,  $n = 1, \dots, N$ , are the **importance weights**
- only constraint:  $\pi(\mathbf{x})$  must be evaluated
- Unfortunately, sometimes we have only access to  $\gamma(\mathbf{x})$ , where  $\pi(\mathbf{x}) = \frac{\gamma(\mathbf{x})}{Z}$ ,  $Z$  is the normalizing constant:
  - basic IS estimator is not possible  $\Rightarrow$  self-normalized estimator:

$$\tilde{I}_N(f) = \sum_{n=1}^N f(\mathbf{x}^{(n)}) w^{(n)}, \quad \mathbf{x}^{(n)} \sim q(\mathbf{x})$$

where  $w^{(n)} = \frac{W^{(n)}}{\sum_{j=1}^N W^{(j)}}$  are the normalized weights ( $\sum_{j=1}^N w^{(j)} = 1$ ).

- \* here the weights  $W^{(n)}$  can be computed by evaluating  $\gamma(\mathbf{x})$  instead.
- Approximation of the targeted distribution

$$\pi(\mathbf{x}) \approx \pi_{\text{IS}}^N(\mathbf{x}) = \sum_{n=1}^N w^{(n)} \delta_{\mathbf{x}^{(n)}}(\mathbf{x})$$

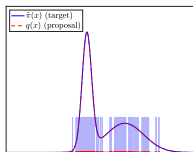
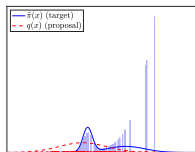
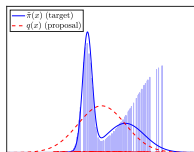


## The variance in IS and the need of better proposals

- A **good proposal**  $q(\mathbf{x})$  is key for the efficiency of IS.
- Variance of the UIS estimator of  $I = \int f(\mathbf{x})\pi(\mathbf{x})d\mathbf{x}$ :

$$\text{Var}_{\pi(\mathbf{x})}(\hat{I}) = \frac{1}{N} \int \frac{f^2(\mathbf{x})\pi^2(\mathbf{x})}{q(\mathbf{x})} d\mathbf{x} - \frac{I^2}{N}$$

- optimal UIS proposal:  $q(\mathbf{x}) \propto |f(\mathbf{x})|\pi(\mathbf{x})$
- for a generic  $f(\mathbf{x})$ ,  $q(\mathbf{x})$  should be as *close* as possible to  $\pi(\mathbf{x})$



- Very difficult to find a good  $q(\mathbf{x})$  a priori:
  - $\pi(\mathbf{x})$  can be only evaluated (up to a normalizing constant)
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- A posteriori metric:  $\widehat{\text{ESS}} = \frac{1}{\sum_{n=1}^N \bar{w}_n^2}$ , although it has known issues<sup>3</sup>
- Use **multiple proposals** (**MIS**) and **explore** the space (**AIS**).

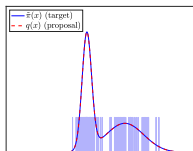
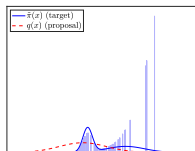
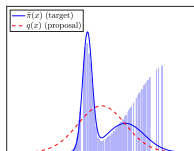
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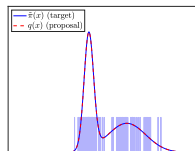
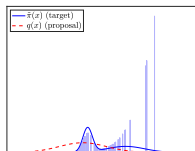
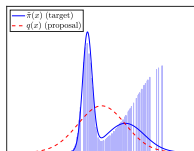
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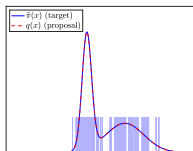
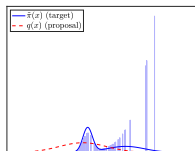
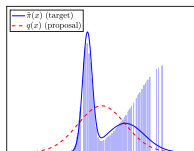
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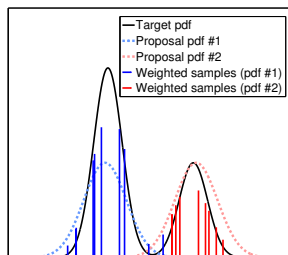
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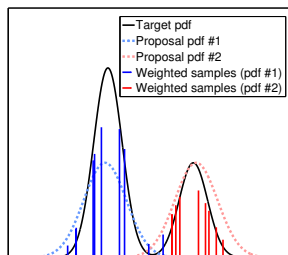
- Set of  $N$  available proposal pdfs  $\{q_n(\mathbf{x})\}_{n=1}^N$ .
  - Example  $N = 2$ :



- For simplicity, we simulate  $N$  samples in total from the set  $\{q_n(\mathbf{x})\}_{n=1}^N$ , but **how?**
  1. **Sampling:**  $\mathbf{x}_n \sim ?$ ,  $n = 1, \dots, N$ .
  2. **Weighting:**  $W_n = ?$ ,  $n = 1, \dots, N$ .
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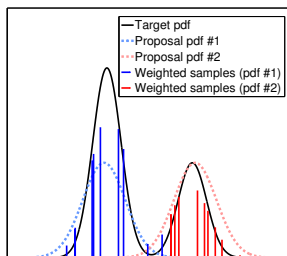
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- Set of  $N$  available proposal pdfs  $\{q_n(\mathbf{x})\}_{n=1}^N$ .
  - Example  $N = 2$ :



- For simplicity, we **simulate**  $N$  **samples** in total from the set  $\{q_n(\mathbf{x})\}_{n=1}^N$ , but **how?**
  1. **Sampling:**  $\mathbf{x}_n \sim ?$ ,  $n = 1, \dots, N$ .
  2. **Weighting:**  $W_n = ?$ ,  $n = 1, \dots, N$ .
- Most known MIS algorithms focus just in the adaptation (AIS):
  - Implement **sampling** and **weighting** in a different way without much justification.



# Multiple Importance Sampling: Interpretations

## a) Population Monte Carlo [Cappe04]:

1. **Sampling:**  $\mathbf{x}_n \sim q_n(\mathbf{x})$ ,  $n = 1, \dots, N$ .
2. **Weighting:**  $W_n = \frac{\pi(\mathbf{x}_n)}{q_n(\mathbf{x}_n)}$ ,  $n = 1, \dots, N$ .
  - Estimators can be unstable.<sup>4</sup>

## b) M-PMC [Douc07a,Cappe08]:

1. **Sampling:**  $\mathbf{x}_n \stackrel{i.i.d}{\sim} \psi(\mathbf{x})$ ,  $n = 1, \dots, N$ .
2. **Weighting:**  $W_n = \frac{\pi(\mathbf{x}_n)}{\psi(\mathbf{x}_n)}$ ,  $n = 1, \dots, N$ .
  - Some proposals can be used more than once, others can be not used.

$$\psi(\mathbf{x}) = \frac{1}{N} \sum_{j=1}^N q_j(\mathbf{x})$$

## c) Deterministic mixture (DM)

1. **Sampling:**  $\mathbf{x}_n \sim q_n(\mathbf{x})$ ,  $n = 1, \dots, N$ .
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  - $\mathbf{x}_n$  is not drawn from  $\psi$ , but estimators are consistent and efficient.

### • Several questions:

- Why all these sampling/weighting schemes are valid?
- Are some schemes better than others?
- Are there other valid schemes?
- Novel theoretical framework for MIS  $\Rightarrow$  Generalized MIS

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<sup>4</sup>O. Cappé, A. Guillin, J.-M. Marin, and C. P. Robert. "Population monte carlo". In: *Journal of Computational and Graphical Statistics* 13.4 (2004), pp: 907–929.

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<sup>5</sup>O. Cappé, R. Douc, A. Guillin, J.-M. Marin, and C. P. Robert. "Adaptive importance sampling in general mixture classes". In: *Statistics and Computing* 18 (2008), pp. 447–459. ↻ 🔍 18/76

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<sup>4</sup>E. Veach and L. J. Guibas. "Optimally combining sampling techniques for Monte Carlo rendering". In: *Proceedings of the 22nd annual conference on Computer graphics and interactive techniques*. 1995, pp. 419–428.

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## Generalized MIS: Schemes with Replacement

- Example with  $N = 3$  proposals. For each  $n \in \{1, 2, 3\}$ :<sup>5</sup>
  - simulate  $j_n \sim \text{Cat}([1, 2, 3]; [1/3, 1/3, 1/3])$
  - simulate  $\mathbf{x}_n \sim q_{j_n}(\mathbf{x})$
  - weight  $W_n = \frac{\pi(\mathbf{x}_n)}{\varphi_n(\mathbf{x}_n)}$  (several options for  $\varphi_n$ )
- 1. and 2. are equivalent to mixture sampling:

$$\mathbf{x}_n \stackrel{i.i.d.}{\sim} \psi(\mathbf{x}) = \frac{1}{3} \sum_{n=1}^N q_n(\mathbf{x})$$

| Available proposals |                |                   |                   |                 |
|---------------------|----------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Sampling            | $j_n$          |                   |                   |                 |
|                     | $\mathbf{x}_n$ | $\mathbf{x}_1 \sim q_3$                                                                            | $\mathbf{x}_2 \sim q_3$                                                                            | $\mathbf{x}_3 \sim q_1$                                                                            |
| Weighting options   | R1             | $\frac{\pi(\mathbf{x}_1)}{q_3(\mathbf{x}_1)}$                                                      | $\frac{\pi(\mathbf{x}_2)}{q_3(\mathbf{x}_2)}$                                                      | $\frac{\pi(\mathbf{x}_3)}{q_1(\mathbf{x}_3)}$                                                      |
|                     | R2             | $\frac{\pi(\mathbf{x}_1)}{\frac{1}{3}(q_3(\mathbf{x}_1) + q_3(\mathbf{x}_1) + q_1(\mathbf{x}_1))}$ | $\frac{\pi(\mathbf{x}_2)}{\frac{1}{3}(q_3(\mathbf{x}_2) + q_3(\mathbf{x}_2) + q_1(\mathbf{x}_2))}$ | $\frac{\pi(\mathbf{x}_3)}{\frac{1}{3}(q_3(\mathbf{x}_3) + q_3(\mathbf{x}_3) + q_1(\mathbf{x}_3))}$ |
|                     | R3             | $\frac{\pi(\mathbf{x}_1)}{\frac{1}{3}(q_1(\mathbf{x}_1) + q_2(\mathbf{x}_1) + q_3(\mathbf{x}_1))}$ | $\frac{\pi(\mathbf{x}_2)}{\frac{1}{3}(q_1(\mathbf{x}_2) + q_2(\mathbf{x}_2) + q_3(\mathbf{x}_2))}$ | $\frac{\pi(\mathbf{x}_3)}{\frac{1}{3}(q_1(\mathbf{x}_3) + q_2(\mathbf{x}_3) + q_3(\mathbf{x}_3))}$ |

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## Generalized MIS: Schemes with No Replacement

- Example with  $N = 3$  proposals. For each  $n \in \{1, 2, 3\}$ :<sup>6</sup>
  1. set  $j_n = n$
  2. simulate  $x_n \sim q_{j_n}(\mathbf{x})$
  3. weight  $W_n = \frac{\pi(\mathbf{x}_n)}{\varphi_n(\mathbf{x}_n)}$  (several options for  $\varphi_n$ )
- 1. and 2. can be seen as mixture sampling  $\mathbf{x}_n \sim \psi(\mathbf{x}) = \frac{1}{3} \sum_{n=1}^N q_n(\mathbf{x})$ 
  - but not i.i.d.!
  - only possible for mixtures with specific weights

| Available proposals |                | ①                                                                                                  | ②                                                                                                  | ③                                                                                                  |
|---------------------|----------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Sampling            | $j_n$          | ①                                                                                                  | ②                                                                                                  | ③                                                                                                  |
|                     | $\mathbf{x}_n$ | $\mathbf{x}_1 \sim q_1$                                                                            | $\mathbf{x}_2 \sim q_2$                                                                            | $\mathbf{x}_3 \sim q_3$                                                                            |
| Weighting options   | N1             | $\frac{\pi(\mathbf{x}_1)}{q_1(\mathbf{x}_1)}$                                                      | $\frac{\pi(\mathbf{x}_2)}{q_2(\mathbf{x}_2)}$                                                      | $\frac{\pi(\mathbf{x}_3)}{q_3(\mathbf{x}_3)}$                                                      |
|                     | N3             | $\frac{\pi(\mathbf{x}_1)}{\frac{1}{3}(q_1(\mathbf{x}_1) + q_2(\mathbf{x}_1) + q_3(\mathbf{x}_1))}$ | $\frac{\pi(\mathbf{x}_2)}{\frac{1}{3}(q_1(\mathbf{x}_2) + q_2(\mathbf{x}_2) + q_3(\mathbf{x}_2))}$ | $\frac{\pi(\mathbf{x}_3)}{\frac{1}{3}(q_1(\mathbf{x}_3) + q_2(\mathbf{x}_3) + q_3(\mathbf{x}_3))}$ |

<sup>6</sup>V. Elvira, L. Martino, D. Luengo, and M. F. Bugallo. “Generalized Multiple Importance Sampling”. In: *Statistical Science* 34.1 (2019), pp. 129–155.

**Theorem.**<sup>7</sup> For any **target distribution**  $\pi(\mathbf{x})$ , any integrable **function**  $h$ , and any **set of proposal densities**  $\{q_n(\mathbf{x})\}_{n=1}^N$  such that the variance of the corresponding MIS estimators is finite,

$$\text{Var}(\hat{I}_{N1}) = \text{Var}(\hat{I}_{R1}) \geq \text{Var}(\hat{I}_{R3}) \geq \text{Var}(\hat{I}_{N3})$$

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## Generalized MIS: Numerical example

- Simulate  $M$  total samples from two proposals:

$$\mathbf{x}_1^{(m)} \sim q_1, \quad m = 1, \dots, M/2$$

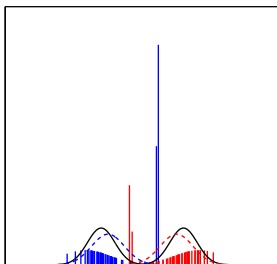
$$\mathbf{x}_2^{(m)} \sim q_2, \quad m = 1, \dots, M/2$$

- Weighting:

(a) **N1**

$$w_1^{(m)} = \frac{\pi(\mathbf{x}_1^{(m)})}{q_1(\mathbf{x}_1^{(m)})}$$

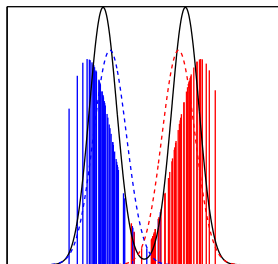
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(b) **N3**

$$w_1^{(m)} = \frac{\pi(\mathbf{x}_1)}{\frac{1}{2}(q_1(\mathbf{x}_1^{(m)}) + q_2(\mathbf{x}_1^{(m)}))}$$

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| MIS Scheme            | R1   | <b>N1</b>   | R2    | N2   | R3   | <b>N3</b>   |
|-----------------------|------|-------------|-------|------|------|-------------|
| $\text{Var}(\hat{Z})$ | 1847 | <b>6874</b> | 10285 | 5474 | 0.01 | <b>0.01</b> |



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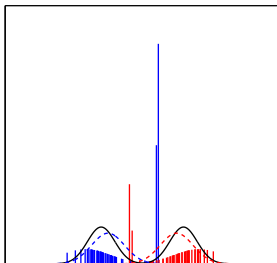
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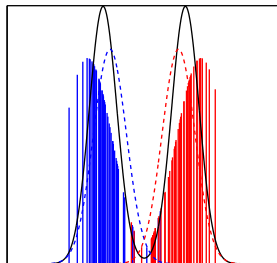
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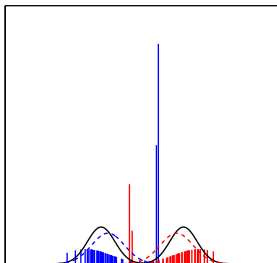
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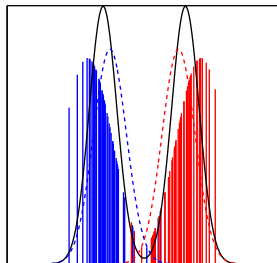
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# Outline

## State-space models and Bayesian filtering

### Importance sampling: basics and advanced methods

- Monte Carlo basics

- Importance sampling (IS) basics

- Multiple importance sampling (MIS)

- Adaptive importance sampling (AIS)**

### Particle filtering

- From IS to sequential IS

- Mini-project

- In search of better particle filters

- Mini-project

## Particle filtering from the MIS and AIS perspectives

### Advanced particle filtering

- PFs in high-dimension spaces

- The choice of  $N$  in PFs

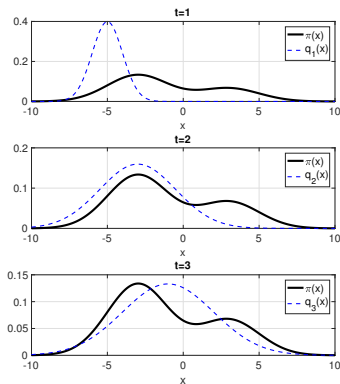
# Adaptive Importance Sampling: Basics

- $N$  proposals  $\{q_{n,j}(\mathbf{x}|\boldsymbol{\theta}_{n,j})\}_{n=1}^N$  adapted over  $j = 1, \dots, J$  iterations

$$\{q_{n,1}(\mathbf{x}|\boldsymbol{\theta}_{n,1})\}_{n=1}^N \rightarrow \{q_{n,2}(\mathbf{x}|\boldsymbol{\theta}_{n,2})\}_{n=1}^N \rightarrow \dots \rightarrow \{q_{n,J}(\mathbf{x}|\boldsymbol{\theta}_{n,J})\}_{n=1}^N$$

- Parametric AIS summarized as:<sup>8</sup>

$$\{\boldsymbol{\theta}_{n,1}\}_{n=1}^N \rightarrow \{\boldsymbol{\theta}_{n,2}\}_{n=1}^N \rightarrow \dots \rightarrow \{\boldsymbol{\theta}_{n,J}\}_{n=1}^N$$



<sup>8</sup>M. F. Bugallo et al. "Adaptive importance sampling: The past, the present, and the future". In: *IEEE Signal Processing Magazine* 34.4 (2017), pp. 60–79.

# Adaptive Importance Sampling: Generic Algorithm

**Initialization:** Choose  $J$ ,  $N$ ,  $K$ ,  $\{q_{n,1}\}_{n=1}^N$ , and initial parameters  $\{\theta_{n,1}\}_{n=1}^N$

For  $j = 1, \dots, J$ :

1. **Sampling:** Simulate  $NK$  samples as

$$\mathbf{x}_{n,j}^{(k)} \sim q_{n,j}(\mathbf{x}|\theta_{n,j}), \quad k = 1, \dots, K, \quad n = 1, \dots, N.$$

2. **Weighting:** Weight the  $NK$  samples as [Elvira19]

$$W_{n,j}^{(k)} = \frac{\pi(\mathbf{x}_{n,j}^{(k)})}{\varphi_{n,j}(\mathbf{x}_{n,j}^{(k)})}, \quad k = 1, \dots, K, \quad n = 1, \dots, N.$$

3. **Adaptation of the parameters:** Update the proposal parameters

$$\{\theta_{n,j}\}_{n=1}^N \xrightarrow{\text{Adapt}} \{\theta_{n,j+1}\}_{n=1}^N$$

**Outputs:**  $NKJ$  weighted samples,  $\{\mathbf{x}_{n,j}^{(k)}, W_{n,j}^{(k)}\}_{n=1, k=1, j=1}^{N, K, J}$

two questions: (1) **Weighting scheme?**<sup>9</sup> (2) **Adaptive procedure of  $\theta_{n,j}$ ?**<sup>10</sup>

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$$\{\theta_{n,j}\}_{n=1}^N \xrightarrow{\text{Adapt}} \{\theta_{n,j+1}\}_{n=1}^N$$

**Outputs:**  $NKJ$  weighted samples,  $\{\mathbf{x}_{n,j}^{(k)}, W_{n,j}^{(k)}\}_{n=1, k=1, j=1}^{N, K, J}$

two questions: (1) Weighting scheme?<sup>9</sup> (2) Adaptive procedure of  $\theta_{n,j}$ ?<sup>10</sup>

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<sup>9</sup>V. Elvira, L. Martino, D. Luengo, and M. F. Bugallo. "Generalized Multiple Importance Sampling". In: *Statistical Science* 34.1 (2019), pp. 129–155.

<sup>10</sup>M. F. Bugallo et al. "Adaptive importance sampling: The past, the present, and the future". In: *IEEE Signal Processing Magazine* 34.4 (2017), pp. 60–75.



# Adaptive Importance Sampling: Generic Algorithm

**Initialization:** Choose  $J$ ,  $N$ ,  $K$ ,  $\{q_{n,1}\}_{n=1}^N$ , and initial parameters  $\{\theta_{n,1}\}_{n=1}^N$

**For**  $j = 1, \dots, J$ :

1. **Sampling:** Simulate  $NK$  samples as

$$\mathbf{x}_{n,j}^{(k)} \sim q_{n,j}(\mathbf{x}|\theta_{n,j}), \quad k = 1, \dots, K, \quad n = 1, \dots, N.$$

2. **Weighting:** Weight the  $NK$  samples as [Elvira19]

$$W_{n,j}^{(k)} = \frac{\pi(\mathbf{x}_{n,j}^{(k)})}{\varphi_{n,j}(\mathbf{x}_{n,j}^{(k)})}, \quad k = 1, \dots, K, \quad n = 1, \dots, N.$$

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$$\{\theta_{n,j}\}_{n=1}^N \xrightarrow{\text{Adapt}} \{\theta_{n,j+1}\}_{n=1}^N$$

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**two questions:** (1) **Weighting scheme?**<sup>9</sup> (2) **Adaptive procedure of  $\theta_{n,j}$ ?**<sup>10</sup>

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## State-space models and Bayesian filtering

### Importance sampling: basics and advanced methods

- Monte Carlo basics

- Importance sampling (IS) basics

- Multiple importance sampling (MIS)

- Adaptive importance sampling (AIS)

### Particle filtering

- From IS to sequential IS

- Mini-project

- In search of better particle filters

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## Particle filtering from the MIS and AIS perspectives

### Advanced particle filtering

- PFs in high-dimension spaces

- The choice of  $N$  in PFs

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# Importance sampling for sequential inference

- Back to our problem: compute the joint  $p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t})$  and/or filtering  $p(\mathbf{x}_t|\mathbf{y}_{1:t})$ :

$$p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) = p(\mathbf{x}_{1:t}, \mathbf{y}_{1:t})/Z$$

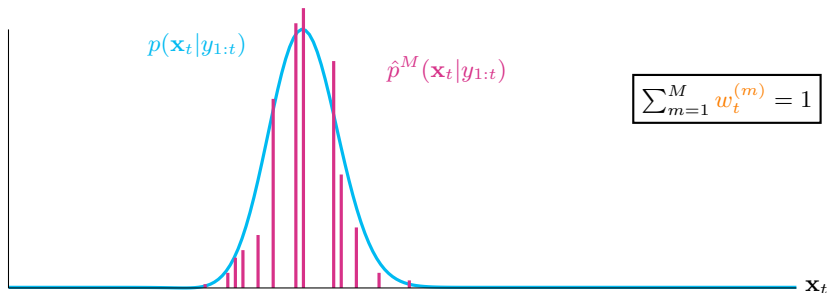
- Norm. constant:  $Z = p(\mathbf{y}_{1:t}) = \int p(\mathbf{x}_{1:t}, \mathbf{y}_{1:t}) d\mathbf{x}_{1:t}$  cannot be computed
  - Marginalization:  $p(\mathbf{x}_t|\mathbf{y}_{1:t}) = \int p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) d\mathbf{x}_{1:t-1}$  cannot be computed
- Importance sampling :
  - Sample  $M$  trajectories  $\mathbf{x}_{1:t}^{(m)} \sim q(\mathbf{x}_{1:t})$ ,  $m = 1, \dots, M$ .
  - Weight each trajectory  $W^{(m)} = \frac{p(\mathbf{x}_{1:t}^{(m)}|\mathbf{y}_{1:t})}{q(\mathbf{x}_{1:t}^{(m)})} \propto \frac{p(\mathbf{x}_{1:t}^{(m)}, \mathbf{y}_{1:t})}{q(\mathbf{x}_{1:t}^{(m)})}$ ,  $m = 1, \dots, M$ .
- Approximate the joint target with weighted trajectories

$$p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) = \sum_{m=1}^M w^{(m)} \delta_{\mathbf{x}_{1:t}^{(m)}}(\mathbf{x}_{1:t})$$

where  $w^{(m)} = \frac{W^{(m)}}{\sum_{j=1}^M W^{(j)}}$  are the normalized weights ( $\sum_{j=1}^M w^{(j)} = 1$ ).

# Particle filtering/sequential Monte Carlo

- Particle filtering/sequential Monte Carlo
- The distributions are approximated by a random measure of  $M$  particles and associated normalized weights  $\mathcal{X} = \{\mathbf{x}_t^{(m)}, w_t^{(m)}\}_{m=1}^M$ 
  - $p(\mathbf{x}_t | \mathbf{y}_{1:t}) \approx \hat{p}^M(\mathbf{x}_t | \mathbf{y}_{1:t}) = \sum_{m=1}^M w_t^{(m)} \delta_{\mathbf{x}_t^{(m)}}(\mathbf{x}_t)$



## Batch importance sampling

- Which proposal  $q(\mathbf{x}_{1:t})$  for sampling/simulating the  $M$  trajectories  $\mathbf{x}_{1:t}^{(n)}$ ?
  - Easiest choice:  $q(\mathbf{x}_{1:t}) = p(\mathbf{x}_{1:t}) \equiv p(\mathbf{x}_1) \cdot p(\mathbf{x}_2|\mathbf{x}_1) \cdot \dots \cdot p(\mathbf{x}_t|\mathbf{x}_{t-1})$ 
    - \* proposal factorizes (it is a choice)
- Batch procedure of IS with  $M$  samples:

1: **for**  $m = 1$  to  $M$  **do**

2:   1. Sample the  $m$ -th trajectory (**sampling**):

$$\mathbf{x}_1^{(m)} \sim p(\mathbf{x}_1)$$

$$\mathbf{x}_2^{(m)} \sim p(\mathbf{x}_2|\mathbf{x}_1^{(m)})$$

...

3:    $\mathbf{x}_{1:t}^{(m)} = [\mathbf{x}_1^{(m)}, \dots, \mathbf{x}_t^{(m)}]$     $\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t|\mathbf{x}_{t-1}^{(m)})$

4:   2. Weight for the  $m$ -th trajectory (**weighting**):

$$W_t^{(m)} = \frac{p(\mathbf{x}_{1:t}^{(m)}, \mathbf{y}_{1:t})}{q(\mathbf{x}_{1:t}^{(m)})} = \frac{p(\mathbf{y}_{1:t}|\mathbf{x}_{1:t}^{(m)})p(\mathbf{x}_{1:t}^{(m)})}{p(\mathbf{x}_{1:t}^{(m)})} = p(\mathbf{y}_{1:t}|\mathbf{x}_{1:t}^{(m)})$$

(likelihood evaluated at the  $m$ -th trajectory)

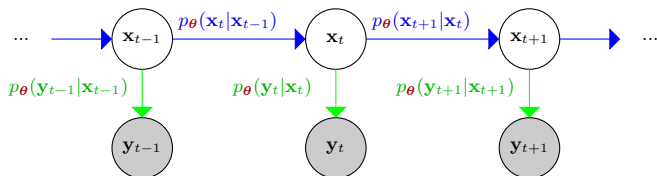
5:   Normalize weights as  $w^{(m)} = \frac{W_t^{(m)}}{\sum_{j=1}^M W_t^{(j)}}$

6: **end for**

$$p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) = \sum_{m=1}^M w^{(m)} \delta_{\mathbf{x}_{1:t}^{(m)}}(\mathbf{x}_{1:t})$$

$$p(\mathbf{x}_t|\mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_t|\mathbf{y}_{1:t}) = \sum_{m=1}^M w^{(m)} \delta_{\mathbf{x}_t^{(m)}}(\mathbf{x}_t) \quad (\text{Monte Carlo marginalization})$$

# Sequential importance sampling (SIS)



- Due to model structure, joint likelihood factorizes as

$$w_t^{(m)} \propto p(\mathbf{y}_{1:t}|\mathbf{x}_{1:t}^{(m)}) = p(\mathbf{y}_1|\mathbf{x}_1^{(m)}) \cdot p(\mathbf{y}_2|\mathbf{x}_2^{(m)}) \cdot \dots \cdot p(\mathbf{y}_t|\mathbf{x}_t^{(m)})$$

- Towards sequential processing:
  - If we receive  $\mathbf{y}_{t+1}$ , can we approximate  $p(\mathbf{x}_{1:t+1}|\mathbf{y}_{1:t+1})$  without re-processing  $\mathbf{y}_{1:t}$ ?

$$\begin{aligned} p(\mathbf{x}_{1:t+1}|\mathbf{y}_{1:t+1}) &= \frac{p(\mathbf{y}_{t+1}, \mathbf{x}_{1:t+1}|\mathbf{y}_{1:t})}{p(\mathbf{y}_{t+1}|\mathbf{y}_{1:t})} = \frac{p(\mathbf{y}_{t+1}|\mathbf{x}_{1:t+1}, \mathbf{y}_{1:t})}{p(\mathbf{y}_{t+1}|\mathbf{y}_{1:t})} p(\mathbf{x}_{1:t+1}|\mathbf{y}_{1:t}) \quad (\text{Bayes}) \\ &= \frac{p(\mathbf{y}_{t+1}|\mathbf{x}_{1:t+1}, \mathbf{y}_{1:t})}{p(\mathbf{y}_{t+1}|\mathbf{y}_{1:t})} p(\mathbf{x}_{t+1}|\mathbf{x}_{1:t}, \mathbf{y}_{1:t}) p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) \quad (\text{factorization}) \\ &= \frac{p(\mathbf{y}_{t+1}|\mathbf{x}_{t+1}) p(\mathbf{x}_{t+1}|\mathbf{x}_t)}{p(\mathbf{y}_{t+1}|\mathbf{y}_{1:t})} p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) \quad (\text{model structure}) \end{aligned}$$

# Sequential importance sampling (SIS)

- Sequential importance sampling (SIS):
    - sample and weight sequentially at time  $t$
    - process each observation  $\mathbf{y}_t$  without reprocessing  $\mathbf{y}_{1:t-1}$
- for**  $t = 2$  to  $T$  **do**
  - for**  $m = 1$  to  $M$  **do**
  1. Sample the  $m$ -th trajectory at time  $t$ :

$$\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(m)})$$

- $\mathbf{x}_{1:t}^{(m)} = [\mathbf{x}_{1:t-1}^{(m)}, \mathbf{x}_t^{(m)}]$

2. Weight for the  $m$ -th trajectory:

$$W_t^{(m)} = p(\mathbf{y}_{1:t} | \mathbf{x}_{1:t}^{(m)}) = W_{t-1}^{(m)} p(\mathbf{y}_t | \mathbf{x}_t^{(m)})$$

- end for**

- end for**

8. Normalize weights:  $w_T^{(m)} = \frac{W_T^{(m)}}{\sum_{j=1}^M W_T^{(j)}}$ ,  $m = 1, \dots, M$ .

9. **return**  $M$  trajectories with their weights:  $\{\mathbf{x}_{1:T}^{(m)}, w_T^{(m)}\}_{m=1}^M$



## Quality of the approximation and resampling step

- We can approximate any function  $f$  of the desired  $p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t})$  with the self-normalized estimator:

$$\tilde{I}_N(f) = \sum_{m=1}^M f(\mathbf{x}_{1:t}^{(m)}) w_t^{(m)}$$

- Example: mean of the filtering distribution  $p(\mathbf{x}_t|\mathbf{y}_{1:t})$  is the weighted average of the particles at time  $t$ .

$$\tilde{I}_N = \sum_{m=1}^M \mathbf{x}_t^{(m)} w_t^{(m)}$$

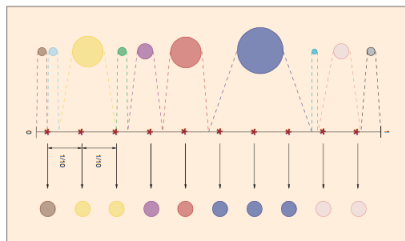
- The quality of the approximation depends on weights variability:
  - ★ if one particle gets a weight  $\approx 1$  (and the others almost zero), the approximation is **very bad**.

- **Resampling step:**

- at each time  $t$  we **kill bad trajectories** with very **low weight** and **replicate good trajectories** (before processing future observations)
- implicit improvement of the marginal proposal for  $t + 1$

## Resampling step

- Resampling step: third step (after 1. Sampling, and 2. Weighting)
  - easy but **necessary** to make the PF work.
- At time  $t$ , after computing the weights we **replicate good particles** and **kill bad particles**.
  - Urn example: draw i.i.d.  $M$  balls (particles) with replacement, with probability equal to the associated weights.
  - More precisely:  $\mathcal{X}_t = \{\mathbf{x}_t^{(m)}, w_t^{(m)}\}_{m=1}^M$  forms an empirical distribution  $p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) \equiv \sum_{j=1}^M w_t^{(j)} \delta_{\mathbf{x}_t^{(j)}}(\mathbf{x})$ .
- \* Sample  $M$  particles,  $\tilde{\mathbf{x}}_t^{(m)} \sim p^M(\mathbf{x}_t | \mathbf{y}_{1:t})$ ,  $m = 1, \dots, M$ .



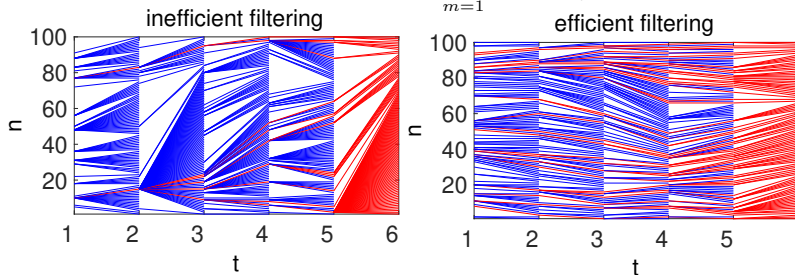
(Source in <sup>11</sup>)

<sup>11</sup>P. M. Djuric et al. "Particle filtering". In: *IEEE signal processing magazine* 20.5 (2003), pp. 19–38.

## Side effect of the resampling step

- Resampling:
  - reduces particle degeneracy (few samples get most of the probability mass)
  - side effect of introducing path degeneracy (few ancestor particles surviving)
- recall we approximate the joint target with weighted trajectories

$$p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) = \sum_{m=1}^M w^{(m)} \delta_{\mathbf{x}_{1:t}^{(m)}}(\mathbf{x}_{1:t})$$



- Resampling only when particle degeneracy is severe:
  - Effective sampling size (ESS):<sup>12</sup>  $\widehat{\text{ESS}} = \frac{1}{\sum_{m=1}^M [w_t^{(m)}]^2}$
  - Adaptive resampling: resample only if  $\widehat{\text{ESS}} < \gamma$ , with  $1 \leq \gamma \leq M$ 
    - \*  $\gamma = M$ : resample always (equiv. BPF)
    - \*  $\gamma = 1$ : never resample (equiv. batch IS)

<sup>12</sup>V. Elvira, L. Martino, and C. P. Robert. “Rethinking the effective sample size”. In: *International Statistical Review* 90.3 (2022), pp. 525–550.

## The bootstrap PF (BPF)

- Bootstrap PF  $\equiv$  Sequential importance sampling resampling (SISR)<sup>13</sup>

(i) Initialization. At time  $t = 0$ ,  $\tilde{\mathbf{x}}_0^{(m)} \sim p(\mathbf{x}_0)$ ,  $m = 1, \dots, M$ .

(ii) Recursive step. At time  $t$ ,

1. **Prediction** (particles propagation):  $\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$

2. **Update** (weights calculation): compute the normalized weights as  $w_t^{(m)} \propto p(\mathbf{y}_t | \mathbf{x}_t^{(m)})$  associated to trajectory  $\mathbf{x}_{1:t}^{(m)} = [\tilde{\mathbf{x}}_{1:t-1}^{(m)}, \mathbf{x}_t^{(m)}]$

3. **Multinomial resampling**

a) simulate  $i^{(m)} \sim \text{Cat}([1, \dots, M]; [w_t^{(1)}, \dots, w_t^{(M)}])$ ,  $m = 1, \dots, M$

b) set  $\tilde{\mathbf{x}}_{1:t}^{(m)} = \mathbf{x}_{1:t}^{(i^{(m)})}$ ,  $m = 1, \dots, M$

equivalent to simulate  $M$  i.i.d. samples from the approx. filtering dist.

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<sup>13</sup>N. J. Gordon, D. J. Salmond, and A. F. Smith. "Novel approach to nonlinear/non-Gaussian Bayesian state estimation". In: *IEE proceedings F (radar and signal processing)*. Vol. 140. 2. IET. 1993, pp. 107–113.

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$$\tilde{\mathbf{x}}_t^{(m)} \sim p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) \equiv \sum_{j=1}^M w_t^{(j)} \delta_{\mathbf{x}_t^{(j)}}(\mathbf{x})$$

$$p(\mathbf{x}_{1:t} | \mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_{1:t} | \mathbf{y}_{1:t}) = \sum_{m=1}^M w_t^{(m)} \delta_{\mathbf{x}_{1:t}^{(m)}}(\mathbf{x}_{1:t})$$

$$p(\mathbf{x}_t | \mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) = \sum_{m=1}^M w_t^{(m)} \delta_{\mathbf{x}_t^{(m)}}(\mathbf{x}_t) \approx \frac{1}{M} \sum_{m=1}^M \delta_{\tilde{\mathbf{x}}_t^{(m)}}(\mathbf{x}_t)$$

<sup>13</sup>N. J. Gordon, D. J. Salmond, and A. F. Smith. "Novel approach to nonlinear/non-Gaussian Bayesian state estimation". In: *IEE proceedings F (radar and signal processing)*. Vol. 140. 2. IET. 1993, pp. 107–113.



# The bootstrap PF (BPF)

- Bootstrap PF  $\equiv$  Sequential importance sampling resampling (SISR)<sup>13</sup>

(i) Initialization. At time  $t = 0$ ,  $\tilde{\mathbf{x}}_0^{(m)} \sim p(\mathbf{x}_0)$ ,  $m = 1, \dots, M$ .

(ii) Recursive step. At time  $t$ ,

1. **Prediction** (particles propagation):  $\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$

2. **Update** (weights calculation): compute the normalized weights as  $w_t^{(m)} \propto p(\mathbf{y}_t | \mathbf{x}_t^{(m)})$  associated to trajectory  $\mathbf{x}_{1:t}^{(m)} = [\tilde{\mathbf{x}}_{1:t-1}^{(m)}, \mathbf{x}_t^{(m)}]$

3. **Multinomial resampling**

a) simulate  $i^{(m)} \sim \text{Cat}([1, \dots, M]; [w_t^{(1)}, \dots, w_t^{(M)}])$ ,  $m = 1, \dots, M$

b) set  $\tilde{\mathbf{x}}_{1:t}^{(m)} = \mathbf{x}_{1:t}^{(i^{(m)})}$   $m = 1, \dots, M$

equivalent to simulate  $M$  i.i.d. samples from the approx. filtering dist.

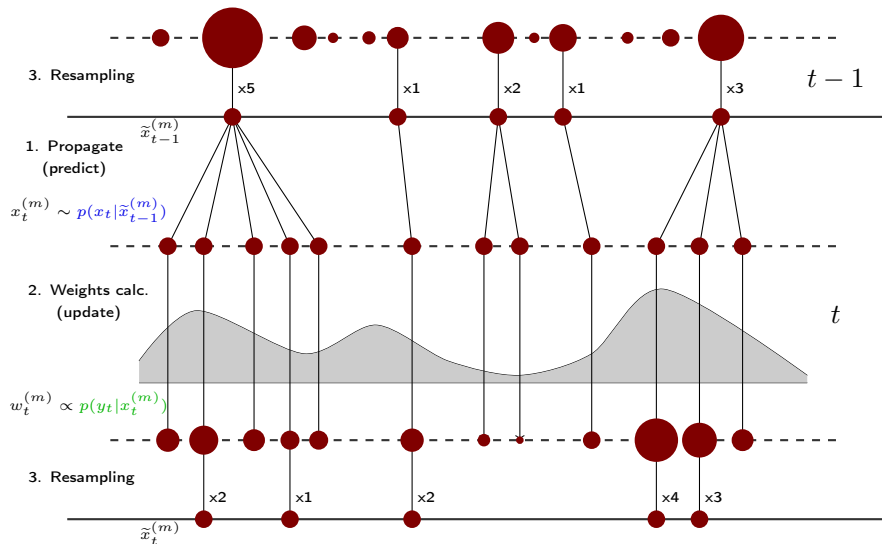
$$\tilde{\mathbf{x}}_t^{(m)} \sim p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) \equiv \sum_{j=1}^M w_t^{(j)} \delta_{\mathbf{x}_t^{(j)}}(\mathbf{x})$$

$$p(\mathbf{x}_{1:t} | \mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_{1:t} | \mathbf{y}_{1:t}) = \sum_{m=1}^M w_t^{(m)} \delta_{\mathbf{x}_{1:t}^{(m)}}(\mathbf{x}_{1:t})$$

$$p(\mathbf{x}_t | \mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) = \sum_{m=1}^M w_t^{(m)} \delta_{\mathbf{x}_t^{(m)}}(\mathbf{x}_t) \approx \frac{1}{M} \sum_{m=1}^M \delta_{\tilde{\mathbf{x}}_t^{(m)}}(\mathbf{x}_t)$$

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# Bootstrap particle filter



# Outline

## State-space models and Bayesian filtering

### Importance sampling: basics and advanced methods

- Monte Carlo basics

- Importance sampling (IS) basics

- Multiple importance sampling (MIS)

- Adaptive importance sampling (AIS)

### Particle filtering

- From IS to sequential IS

#### **Mini-project**

- In search of better particle filters

- Mini-project

## Particle filtering from the MIS and AIS perspectives

### Advanced particle filtering

- PFs in high-dimension spaces

- The choice of  $N$  in PFs

## Mini-project: BF for the Lorenz 63 model

- Take the same state-space model as for the Kalman filtering problem (stochastic Lorenz 63, with Euler discretisation and nonlinear observations) and code a standard bootstrap filter with  $N$  particles.
- Compare the performance of the bootstrap filter and the non-linear extensions of KF (e.g., CKF): try different values of state noise variance and observational noise variance, gap between observations, and  $M$  (number of particles).

# Outline

## State-space models and Bayesian filtering

### Importance sampling: basics and advanced methods

- Monte Carlo basics

- Importance sampling (IS) basics

- Multiple importance sampling (MIS)

- Adaptive importance sampling (AIS)

### Particle filtering

- From IS to sequential IS

- Mini-project

- In search of better particle filters**

- Mini-project

## Particle filtering from the MIS and AIS perspectives

### Advanced particle filtering

- PFs in high-dimension spaces

- The choice of  $N$  in PFs

## Beyond BPF

- The  $m$ -th proposal in BPF is the transition kernel  $q(\mathbf{x}_t) = p(\mathbf{x}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$ 
  - note that is conditioned on  $\tilde{\mathbf{x}}_{t-1}^{(m)}$  which comes from a resampling step.
  - proposal of all samples can be interpreted as approximate predictive, since resampling  $(t-1)$  + propagation at  $t$  = mixture sampling:

$$\mathbf{x}_t^{(m)} \stackrel{i.i.d.}{\sim} p^M(\mathbf{x}_t | \mathbf{y}_{1:t-1}) = \sum_{m=1}^M w_t^{(m)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(m)})$$

- From IS theory, high variance of the importance weights  $\Rightarrow$  low efficiency/accuracy of the filter
- **Intuition** (imprecise): inefficiency in BPF will happen when predictive  $p(\mathbf{x}_t | \mathbf{y}_{1:t-1})$ , thus mixture proposal, differs from **filtering distribution**:

$$p(\mathbf{x}_t | \mathbf{y}_{1:t}) = \frac{p(\mathbf{y}_t | \mathbf{x}_t) p(\mathbf{x}_t | \mathbf{y}_{1:t-1})}{p(\mathbf{y}_t | \mathbf{y}_{1:t-1})}$$

- equivalent to say that  $\mathbf{y}_t$  is **informative**, which happens when:
  - \*  $p(\mathbf{y}_t | \mathbf{x}_t)$  is "peaky" compared to  $p(\mathbf{x}_t | \mathbf{y}_{1:t-1})$  (e.g., very low-variance observation noise)
  - \*  $p(\mathbf{y}_t | \mathbf{x}_t)$  is placed in a "different" area of the space compared to  $p(\mathbf{x}_t | \mathbf{y}_{1:t-1})$  (e.g., outlier observation)
- in those scenarios, great variability of the weights.

## An adapted PF with generic proposal

- A good proposal must include knowledge about  $\mathbf{y}_t$ 
  - avoid particles being sampled into regions of the state space which are unlikely in light of that observation
- Generic sampling:

$$\mathbf{x}_{1:t}^{(m)} \sim q(\mathbf{x}_{1:t}) = \prod_{k=1}^t q(\mathbf{x}_k; \mathbf{x}_{1:k-1}, \mathbf{y}_{1:k})$$

with trajectory weight:

$$W_t^{(m)} = \frac{p(\mathbf{x}_{1:t}^{(m)}, \mathbf{y}_{1:t})}{q(\mathbf{x}_{1:t}^{(m)})} = \frac{p(\mathbf{y}_{1:t} | \mathbf{x}_{1:t}^{(m)}) p(\mathbf{x}_{1:t}^{(m)})}{q(\mathbf{x}_{1:t}^{(m)})} = \frac{\prod_{k=1}^t p(\mathbf{y}_k | \mathbf{x}_k^{(m)}) p(\mathbf{x}_k^{(m)} | \mathbf{x}_{k-1}^{(m)})}{\prod_{k=1}^t q(\mathbf{x}_k^{(m)}; \mathbf{x}_{1:k-1}^{(m)}, \mathbf{y}_{1:k})}$$

- possible proposal choices:
  - BPF (online-oriented) uninformative:

$$\mathbf{x}_t^{(m)} \sim q(\mathbf{x}_t; \mathbf{x}_{1:t-1}^{(m)}, \mathbf{y}_{1:t}) = q(\mathbf{x}_t; \mathbf{x}_{t-1}^{(m)}) = p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(m)})$$

- still convenient (online-oriented) but more informative:

$$\mathbf{x}_t^{(m)} \sim q(\mathbf{x}_t; \mathbf{x}_{1:t-1}^{(m)}, \mathbf{y}_{1:t}) = q(\mathbf{x}_t; \mathbf{x}_{t-1}^{(m)}, \mathbf{y}_t)$$

# An adapted PF with generic proposal

- (i) Initialization. At time  $t = 0$ ,  $\tilde{\mathbf{x}}_0^{(m)} \sim p(\mathbf{x}_0)$ ,  $m = 1, \dots, M$ .
- (ii) Recursive step. At time  $t$ ,
1. **Sampling/propagation:**  $\mathbf{x}_t^{(m)} \sim q(\mathbf{x}_t; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t)$  associated to trajectory  $\mathbf{x}_{1:t}^{(m)} = [\tilde{\mathbf{x}}_{1:t-1}^{(m)}, \mathbf{x}_t^{(m)}]$
  2. **Weights calculation:** compute the normalized weights as
$$w_t^{(m)} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \tilde{\mathbf{x}}_{t-1}^{(m)})}{q(\mathbf{x}_t^{(m)}; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t)}$$
  3. **Multinomial resampling**
    - a) simulate  $i^{(m)} \sim \text{Cat}([1, \dots, M]; [w_t^{(1)}, \dots, w_t^{(M)}])$ ,  $m = 1, \dots, M$
    - b) set  $\tilde{\mathbf{x}}_t^{(m)} = \mathbf{x}_t^{(i^{(m)})}$ ,  $m = 1, \dots, M$equivalent to simulate  $M$  i.i.d. samples from the approx. filtering dist.

$$\tilde{\mathbf{x}}_t^{(m)} \sim p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) \equiv \sum_{j=1}^M w_t^{(j)} \delta_{\mathbf{x}_t^{(j)}}(\mathbf{x})$$



## An adapted PF with generic proposal

(i) Initialization. At time  $t = 0$ ,  $\tilde{\mathbf{x}}_0^{(m)} \sim p(\mathbf{x}_0)$ ,  $m = 1, \dots, M$ .

(ii) Recursive step. At time  $t$ ,

1. **Sampling/propagation:**  $\mathbf{x}_t^{(m)} \sim q(\mathbf{x}_t; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t)$  associated to trajectory

$$\mathbf{x}_{1:t}^{(m)} = [\tilde{\mathbf{x}}_{1:t-1}^{(m)}, \mathbf{x}_t^{(m)}]$$

2. **Weights calculation:** compute the normalized weights as

$$w_t^{(m)} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \tilde{\mathbf{x}}_{t-1}^{(m)})}{q(\mathbf{x}_t^{(m)}; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t)}$$

3. **Multinomial resampling**

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equivalent to simulate  $M$  i.i.d. samples from the approx. filtering dist.

$$\tilde{\mathbf{x}}_t^{(m)} \sim p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) \equiv \sum_{j=1}^M w_t^{(j)} \delta_{\mathbf{x}_t^{(j)}}(\mathbf{x})$$

## An adapted PF with generic proposal

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$$\tilde{\mathbf{x}}_t^{(m)} \sim p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) \equiv \sum_{j=1}^M w_t^{(j)} \delta_{\mathbf{x}_t^{(j)}}(\mathbf{x})$$

# In search of “optimal” filters

- **Proposal choice** to minimize variance of

$$w_t^{(m)} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \tilde{\mathbf{x}}_{t-1}^{(m)})}{q(\mathbf{x}_t^{(m)}; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t)}$$

- “optimal” kernel:  $q(\mathbf{x}_t; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t) = p(\mathbf{x}_t | \mathbf{y}_t, \tilde{\mathbf{x}}_{t-1}^{(m)})$ 
  - \* proportional to the numerator  $p(\mathbf{y}_t, \mathbf{x}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$
  - \* intractable kernel
  - \* reduces the variance of each weight (constant) but still weights are different across them:

$$W_t^{(m)} = p(\mathbf{y}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$$

- Even if available, the kernel does not solve all the problems: the resampling remains blind to the new observation  $\mathbf{y}_t$ 
  - **Goal:** we would like to modify the resampling weights to replicate trajectories at  $t - 1$  that will perform better in  $t$

## In search of “optimal” filters: trajectory perspective

- Recap: BPF and adapted PF proceed in the following order:
  - Resampling ( $t - 1$ ): resample trajectories  $\tilde{\mathbf{x}}_{1:t-1}^{(n)}$  ( $\mathbf{y}_t$  is **not used** in adapted PF nor in BPF)
    - ★ recall: equivalent to simulating at  $t - 1$  as
$$\tilde{\mathbf{x}}_{1:t-1}^{(m)} \sim p^M(\mathbf{x}_{1:t-1} | \mathbf{y}_{1:t-1}) \equiv \sum_{j=1}^M w_{t-1}^{(j)} \delta_{\mathbf{x}_{1:t-1}^{(j)}}(\mathbf{x}_{1:t-1})$$
  - Sampling ( $t$ ): propagate from  $\tilde{\mathbf{x}}_{t-1}^{(m)}$  to  $\mathbf{x}_t^{(m)}$  ( $\mathbf{y}_t$  is **used in adapted PF** but **not used in BPF**)
  - Weighting ( $t$ ): Bayesian update ( $\mathbf{y}_t$  is used)
- Is it possible to use  $\mathbf{y}_t$  also at resampling?

| Filter       | Sampling | Resampling |
|--------------|----------|------------|
| Bootstrap PF | ✗        | ✗          |
| Adapted PF   | ✓        | ✗          |
| Optimal PF   | ✓        | ✓          |

## In search of better filters: trajectory perspective

- More precisely, resample trajectories from  $p^M(\mathbf{x}_{1:t-1}|\mathbf{y}_{1:t})$  instead of  $p^M(\mathbf{x}_{1:t-1}|\mathbf{y}_{1:t-1})$ ?

$$\begin{aligned}
 p(\mathbf{x}_{1:t-1}|\mathbf{y}_{1:t}) &= \int p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) d\mathbf{x}_t \\
 &= \int p(\mathbf{y}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{x}_{t-1})p(\mathbf{x}_{1:t-1}|\mathbf{y}_{1:t-1})d\mathbf{x}_t \\
 &= p(\mathbf{x}_{1:t-1}|\mathbf{y}_{1:t-1}) \int p(\mathbf{y}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{x}_{t-1})d\mathbf{x}_t \\
 &\approx \left[ \sum_{j=1}^M w_{t-1}^{(j)} \delta_{\mathbf{x}_{1:t-1}^{(j)}}(\mathbf{x}_{1:t-1}) \right] \int p(\mathbf{y}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{x}_{t-1})d\mathbf{x}_t \\
 &= \sum_{j=1}^M w_{t-1}^{(j)} \delta_{\mathbf{x}_{1:t-1}^{(j)}}(\mathbf{x}_{1:t-1}) \underbrace{\int p(\mathbf{y}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{x}_{t-1}^{(j)})d\mathbf{x}_t}_{v_t^{(j)}=p(\mathbf{y}_t|\mathbf{x}_{t-1}^{(j)})} \\
 &= \sum_{j=1}^M \underbrace{w_{t-1}^{(j)} v_t^{(j)}}_{\lambda_t^{(j)}} \delta_{\mathbf{x}_{1:t-1}^{(j)}}(\mathbf{x}_{1:t-1})
 \end{aligned}$$

- $v_t^{(j)}$  is intractable  $\Rightarrow$  cheap approximation:  $v_t^{(j)} \approx p(\mathbf{y}_t|\bar{\mathbf{x}}_t^{(j)})$ 
  - justification:  $p(\mathbf{x}_t|\mathbf{x}_{t-1}^{(j)}) \approx \delta_{\bar{\mathbf{x}}_t^{(j)}}(\mathbf{x}_t)$ , where  $\bar{\mathbf{x}}_t^{(j)} = \mathbb{E}_{p(\mathbf{x}_t|\mathbf{x}_{t-1}^{(j)})}[\mathbf{x}_t]$



## In search of better filters: trajectory perspective

New approach:

1. Resampling ( $t-1$ ): resample trajectories  $\tilde{\mathbf{x}}_{1:t-1}^{(m)}$  ( $\mathbf{y}_t$  is now **used**)
2. Sampling ( $t$ ): simulate from  $q(\mathbf{x}_t; \tilde{\mathbf{x}}_{t-1}, \mathbf{y}_t)$  ( $\mathbf{y}_t$  potentially **used**)
  - trajectory proposal is now

$$q(\mathbf{x}_t; \mathbf{x}_{1:t-1}) = q(\mathbf{x}_t; \mathbf{x}_{t-1}, \mathbf{y}_t) \sum_{j=1}^M \lambda_t^{(j)} \delta_{\mathbf{x}_{1:t-1}^{(j)}}(\mathbf{x}_{1:t-1})$$

and sampling as

a) simulate  $i^{(m)} \sim \text{Cat}([1, \dots, M]; [\lambda_t^{(1)}, \dots, \lambda_t^{(M)}])$ ,  $m = 1, \dots, M$

b) simulate  $\mathbf{x}_t^{(m)} \sim q(\mathbf{x}_t; \mathbf{x}_{t-1}^{(i^{(m)})}, \mathbf{y}_t)$ ,  $m = 1, \dots, M$

3. Weighting ( $t$ ) in the joint space, using  $\mathbf{x}_{1:t-1}^{(j)}$  fulfills

$$\begin{aligned} q(\mathbf{x}_{1:t-1}^{(j)}; \mathbf{y}_{1:t}) &= \frac{\lambda_t^{(j)}}{w_t^{(j)}} p^M(\mathbf{x}_{1:t-1}^{(j)} | \mathbf{y}_{1:t-1}), \\ w_t(\mathbf{x}_{1:t}^{(m)}) &= \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^m | \mathbf{x}_{t-1}^{(i^{(m)})}) p(\mathbf{x}_{1:t-1}^{(i^{(m)})} | \mathbf{y}_{1:t-1})}{q(\mathbf{x}_t^{(m)}; \mathbf{x}_{1:t-1}^{(i^{(m)})}, \mathbf{y}_t) q(\mathbf{x}_{1:t-1}^{(i^{(m)})} | \mathbf{y}_{1:t})} \\ &= \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^m | \mathbf{x}_{t-1}^{(i^{(m)})}) p(\mathbf{x}_{1:t-1}^{(i^{(m)})} | \mathbf{y}_{1:t-1})}{q(\mathbf{x}_t^{(m)}; \mathbf{x}_{1:t-1}^{(i^{(m)})}, \mathbf{y}_t) \frac{\lambda_t^{(i^{(m)})}}{w_t^{(i^{(m)})}} p^M(\mathbf{x}_{1:t-1}^{(i^{(m)})} | \mathbf{y}_{1:t-1})} \\ &\approx \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^m | \mathbf{x}_{t-1}^{(i^{(m)})})}{q(\mathbf{x}_t^{(m)}; \mathbf{x}_{1:t-1}^{(i^{(m)})}, \mathbf{y}_t) p(\mathbf{y}_t | \mu_t^{(i^{(m)})})} \end{aligned}$$

## Auxiliary PF (APF)

- Proposed in <sup>14</sup> as an alternative to BPF
  - APF improves sometimes the performance of BPF, but **not always**.
  - it attempts to sample in better areas in light of the new observation  $y_t$
- (i) Initialization. At time  $t = 0$ ,  $\mathbf{x}_0^{(m)} \sim p(\mathbf{x}_0)$ , and  $w_0^{(m)} = 1/M$ ,  $m = 1, \dots, M$ .
- (ii) Recursive step. At time  $t > 0$ ,
  - Modify weights before resampling.** Compute

$$\bar{\mathbf{x}}_t^{(m)} = \mathbb{E}_{p(\mathbf{x}_t|\mathbf{x}_{t-1}^{(m)})}[\mathbf{x}_t], \quad m = 1, \dots, M.$$

and the normalized weights ( $\sum_{m=1}^M \lambda_t^{(m)} = 1$ )

$$\lambda_t^{(m)} \propto p(\mathbf{y}_t|\bar{\mathbf{x}}_t^{(m)})w_{t-1}^{(m)}, \quad m = 1, \dots, M,$$

- Delayed resampling.** Select the indexes  $i^{(m)} = j$ , with probability proportional to  $\lambda_t^{(j)}$ ,  $m = 1, \dots, M$
- Prediction.**  $\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t|\mathbf{x}_{t-1}^{(i^{(m)})})$ ,  $m = 1, \dots, M$ .
- Update.** Compute the normalized weights as

$$w_t^{(m)} \propto \frac{p(\mathbf{y}_t|\mathbf{x}_t^{(m)})}{p(\mathbf{y}_t|\bar{\mathbf{x}}_t^{(i^{(m)})})}, \quad m = 1, \dots, M.$$

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<sup>14</sup>M. K. Pitt and N. Shephard. "Filtering via simulation: Auxiliary particle filters". In: *Journal of the American statistical association* 94.446 (1999), pp. 590–599. > < ≡ ≡ ≡

# Outline

## State-space models and Bayesian filtering

## Importance sampling: basics and advanced methods

- Monte Carlo basics

- Importance sampling (IS) basics

- Multiple importance sampling (MIS)

- Adaptive importance sampling (AIS)

## Particle filtering

- From IS to sequential IS

- Mini-project

- In search of better particle filters

- Mini-project**

## Particle filtering from the MIS and AIS perspectives

## Advanced particle filtering

- PFs in high-dimension spaces

- The choice of  $N$  in PFs

## Mini-project

- Assume, again, the stochastic Lorenz 63 model, time-discretised using an Euler-Maruyama scheme:

$$X_{1,n} = X_{1,n-1} - hs(X_{1,n-1} - X_{2,n-1}) + \sigma\sqrt{h}Z_{1,n},$$

$$X_{2,n} = X_{2,n-1} + h(rX_{1,n-1} - X_{2,n-1} - X_{1,n-1}X_{3,n-1}) + \sigma\sqrt{h}Z_{2,n},$$

$$X_{3,n} = X_{3,n-1} + h(X_{1,n-1}X_{2,n-1} - bX_{3,n-1}) + \sigma\sqrt{h}Z_{3,n},$$

where  $Z_{i,n} \sim \mathcal{N}(0, 1)$ , the state is  $X_n = [X_{1,n}, X_{2,n}, X_{3,n}]^\top$  and the parameters are  $(s, r, b) = (10, 28, \frac{8}{3})$ . You may 'play around' with the value of  $\sigma$  as in the previous mini-projects (start with  $\sigma = \frac{1}{2}$ ).

- Assume linear observations, namely  $Y_n = X_{1,n} + \sigma_u U_n$ , where  $U_n \sim \mathcal{N}(0, 1)$ , every  $B$  discrete time steps (e.g.,  $B = 40$ ). Again, you may try different values of the parameter  $\sigma_u$  (you may start with  $\sigma_u = 2$ ).

# Mini-project

- For the state space model in the previous slide, code a standard SIR algorithm and a SIR algorithm with a Gaussian proposal.  
Try different configurations of the model noise parameters  $(\sigma_u, \sigma, M)$  and plot the ground-truth signals and their estimates with both filters. Reference values:  
 $\sigma_u = 2, \sigma = 1, B = 40$  and  $M = 50$  particles in the SIR algorithm.  
The simulation should also return estimation errors (e.g., average square errors over time) for the two filters.

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## Particle filtering from the MIS and AIS perspectives

### Advanced particle filtering

- PFs in high-dimension spaces

- The choice of  $N$  in PFs

## Revisiting standard filters from the MIS perspective

- Both BPF and APF (and other filters) use several proposals at each  $t$
- e.g., BPF proposal is  $\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$ 
  - potentially  $M$  different values  $\tilde{\mathbf{x}}_{t-1}^{(m)}$ , i.e.,  $M$  proposals.
  - at least two views:
    1. each sample  $\mathbf{x}_t^{(m)}$  is simulated from  $p(\mathbf{x}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$
    2. all samples are i.i.d. samples from the mixture

$$p^M(\mathbf{x}_t | \mathbf{y}_{1:t-1}) = \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)})$$

- Is it possible to re-interpret BPF and other filters from a MIS perspective?

## A generic particle filtering from the MIS perspective

- (i) Initialization. At time  $t = 0$ ,  $\mathbf{x}_0^{(m)} \sim p(\mathbf{x}_0)$ ,  $w_0^{(m)} = 1/M$ ,  $m = 1, \dots, M$ .
- (ii) Recursive step. At time  $t > 0$ ,

1 **Proposal adaptation/selection.** Select the MIS proposal of the form

$$\psi_t(\mathbf{x}_t) = \sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)}),$$

2 **Sampling.** Draw samples according to

$$\mathbf{x}_t^{(m)} \sim \psi_t(\mathbf{x}_t), \quad m = 1, \dots, M.$$

3 **Weighting.** Compute the normalized IS weights by

$$\begin{aligned} w_t^{(m)} &\propto \frac{p(\mathbf{x}_t^{(m)} | \mathbf{y}_{1:t})}{\psi_t(\mathbf{x}_t^{(m)})} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \mathbf{y}_{1:t-1})}{\psi_t(\mathbf{x}_t^{(m)})} \\ &\approx \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}{\psi_t(\mathbf{x}_t^{(m)})} = \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})} \end{aligned} \quad (2)$$

• Two questions:<sup>15</sup>

1. Selection/adaptation of  $\{\lambda_t^{(j)}\}_{j=1}^M$  to build  $\psi_t(\mathbf{x}_t)$ ?

\* Recall: IS is efficient when  $\psi_t(\mathbf{x}_t)$  is close to  $p(\mathbf{x}_t | \mathbf{y}_{1:t}) \Rightarrow$  AIS

2. Approximate  $w_t^{(m)}$  in (2) to derive BPF, APF. and other/new filters?

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<sup>15</sup>V. Elvira, L. Martino, M. F. Bugallo, and P. M. Djuric. "Elucidating the auxiliary particle filter via multiple importance sampling [lecture notes]". In: *IEEE Signal Processing Magazine* 36.6 (2019), pp. 145–152.



## BPF from the MIS perspective

- (i) Initialization. At time  $t = 0$ ,  $\mathbf{x}_0^{(m)} \sim p(\mathbf{x}_0)$ , and  $w_0^{(m)} = 1/M$ ,  $m = 1, \dots, M$ .
- (ii) Recursive step. At time  $t > 0$ ,
- Proposal adaptation/selection.** Select the MIS proposal of the form

$$\psi_t(\mathbf{x}_t) = \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)}), \quad (\lambda_t^{(j)} = w_{t-1}^{(j)})$$

- Sampling.** Draw samples according to

$$\mathbf{x}_t^{(m)} \sim \psi_t(\mathbf{x}_t), \quad m = 1, \dots, M. \quad (\text{equiv. resampling+propagation})$$

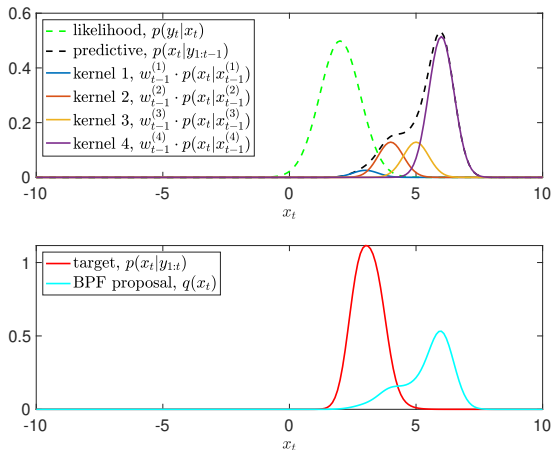
- Weighting.** Compute the normalized IS weights by

$$\begin{aligned} w_t^{(m)} &\propto \frac{p(\mathbf{x}_t^{(m)} | \mathbf{y}_{1:t})}{\psi_t(\mathbf{x}_t^{(m)})} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \mathbf{y}_{1:t-1})}{\psi_t(\mathbf{x}_t^{(m)})} \\ &\approx \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}{\psi_t(\mathbf{x}_t^{(m)})} = \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})} \\ &= p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \end{aligned}$$

- Remark: the BPF proposal matches just the prior of the numerator.<sup>16</sup>

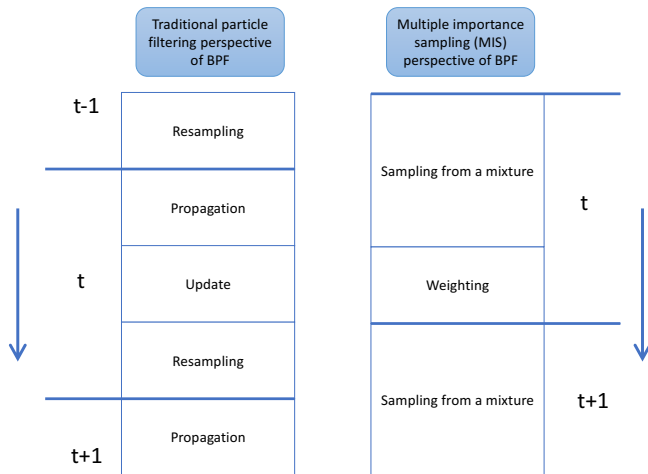
<sup>16</sup>V. Elvira, L. Martino, M. F. Bugallo, and P. M. Djuric. "Elucidating the auxiliary particle filter via multiple importance sampling [lecture notes]". In: *IEEE Signal Processing Magazine* 36.6 (2019), pp. 145–152.

## Toy example: BPF with $M = 4$ particles



- predictive,  $p(x_t|y_{1:t-1}) = \sum_{j=1}^M w_{t-1}^{(j)} p(x_t|x_{t-1}^{(j)})$  with  $w_{t-1} = [0.03, 0.16, 0.16, 0.65]$
- BPF proposal,  $\psi_t^{\text{BPF}}(x_t) = \sum_{j=1}^M \lambda_t^{(j)} p(x_t|x_{t-1}^{(j)})$ , with  $\lambda_t^{\text{BPF}} = w_{t-1}^{(m)} = [0.03, 0.16, 0.16, 0.65]$

# BPF from the MIS perspective



## APF from the MIS perspective

- (i) Initialization. At time  $t = 0$ ,  $\mathbf{x}_0^{(m)} \sim p(\mathbf{x}_0)$ , and  $w_0^{(m)} = 1/M$ ,  $m = 1, \dots, M$ .
- (ii) Recursive step. At time  $t > 0$ ,
- Proposal adaptation/selection.** The weight of each mixture kernel is amplified by the likelihood eval. at its center  $\bar{\mathbf{x}}_t^{(m)} = \mathbb{E}_{p(\mathbf{x}_t|\mathbf{x}_{t-1}^{(m)})}[\mathbf{x}_t]$ , i.e.,

$$\psi_t(\mathbf{x}_t) = \sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)}), \quad \text{with } \lambda_t^{(j)} \propto p(\mathbf{y}_t | \bar{\mathbf{x}}_t^{(j)}) w_{t-1}^{(j)}, \quad j = 1, \dots, M$$

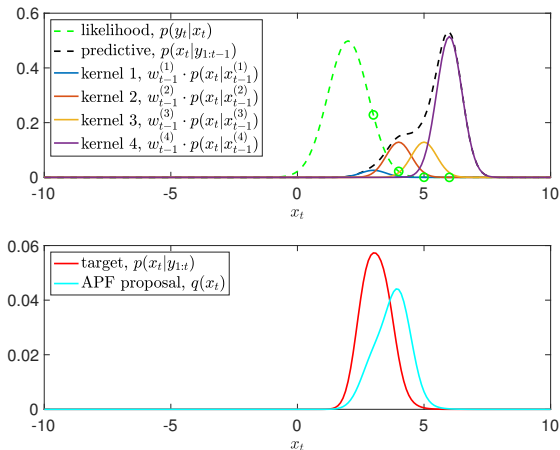
- Sampling.** Draw  $M$  i.i.d. samples from the mixture  $\psi_t(\mathbf{x}_t)$ , i.e.,
  - Select the indexes  $i^{(m)} = j$ , with probability  $\propto \lambda_t^{(j)}$ ,  $m = 1, \dots, M$
  - simulate  $\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(i^{(m)})})$ ,  $m = 1, \dots, M$ .
- Weighting.** Compute the normalized IS weights by

$$\begin{aligned} w_t^{(m)} &\propto \frac{p(\mathbf{x}_t^{(m)} | \mathbf{y}_{1:t})}{\psi_t(\mathbf{x}_t^{(m)})} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \mathbf{y}_{1:t-1})}{\sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})} \approx \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})} \\ &\approx \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) w_{t-1}^{(i^{(m)})} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(i^{(m)})})}{\lambda_t^{(i^{(m)})} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(i^{(m)})})} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) w_{t-1}^{(i^{(m)})} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(i^{(m)})})}{p(\mathbf{y}_t | \bar{\mathbf{x}}_t^{(i^{(m)})}) w_{t-1}^{(i^{(m)})} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(i^{(m)})})} = \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)})}{p(\mathbf{y}_t | \bar{\mathbf{x}}_t^{(i^{(m)})})} \end{aligned}$$

- Remark:<sup>17</sup>
  - implicit assumption: kernels are far apart
  - the APF re-weights the kernels of the prior amplifying them with the likelihood (each of them, independently from the rest).

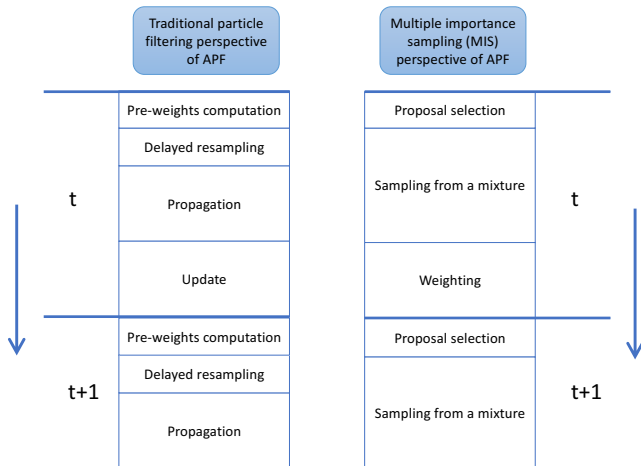
<sup>17</sup>V. Elvira, L. Martino, M. F. Bugallo, and P. M. Djuric. "Elucidating the auxiliary particle filter via multiple importance sampling [lecture notes]". In: *IEEE Signal Processing Magazine* 36.6 (2019), pp. 145–152.

## Toy example: APF with $M = 4$ particles



- predictive,  $p(x_t|y_{1:t-1}) = \sum_{j=1}^M w_{t-1}^{(j)} p(x_t|x_{t-1}^{(j)})$  with  $w_{t-1} = [0.03, 0.16, 0.16, 0.65]$
- APF proposal,  $\psi_t^{\text{APF}}(x_t) = \sum_{j=1}^M \lambda_t^{(j)} p(x_t|x_{t-1}^{(j)})$ , with  $\lambda_t^{\text{APF}} = p(y_t|\bar{x}_t^{(m)}) w_{t-1}^{(m)} = [0.6713, 0.3221, 0.0065, 0.0001]$

## Auxiliary PF (APF) from the MIS perspective



# Improved APF (IAPF)

- IAPF:<sup>18</sup> Based on this MIS interpretation, we improve the APF
  - MIS perspective: the proposal is a **mixture of the same predictive kernels** as in BPF and APF

$$\psi_t(\mathbf{x}_t) = \sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)})$$

with

$$\lambda_t^{(j)} \propto \frac{p(\mathbf{y}_t | \bar{\mathbf{x}}_t^{(j)}) \sum_{k=1}^M w_{t-1}^{(k)} p(\bar{\mathbf{x}}_t^{(j)} | \mathbf{x}_{t-1}^{(k)})}{\sum_{k=1}^M p(\bar{\mathbf{x}}_t^{(j)} | \mathbf{x}_{t-1}^{(k)})}, \quad j = 1, \dots, M.$$

- **Interpretation:** the “amplification”  $\lambda_t^{(j)}$  of  $j$ -th kernel, takes into account where all other kernels are placed (unlike APF)
  - \* if kernels have small overlap,  $\lambda_t^{(j)} \approx p(\mathbf{y}_t | \bar{\mathbf{x}}_t^{(j)}) w_{t-1}^{(j)}$  (IAPF reduces to APF)
- Connection to marginal PFs <sup>19</sup>

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<sup>18</sup>V. Elvira, L. Martino, M. F. Bugallo, and P. M. Djurić. “In search for improved auxiliary particle filters”. In: *2018 26th European Signal Processing Conference (EUSIPCO)*. IEEE, 2018, pp. 1637–1641.

<sup>19</sup>M. Klaas, N. De Freitas, and A. Doucet. “Toward practical N2 Monte Carlo: The marginal particle filter”. In: *arXiv preprint arXiv:1207.1396* (2012).

# Optimized APF (OAPF)

- OAPF:<sup>2021</sup>  $K < M$  kernels form the mixture proposal
  - Optimized  $\lambda$  via non-negative least squares (NNLS) by taking the squared distance between target and mixture proposal at the  $E$  evaluation points

$$\boldsymbol{\lambda}^* = \arg \min_{\boldsymbol{\lambda}} \|\mathbf{Q}\boldsymbol{\lambda} - \tilde{\boldsymbol{\pi}}\|_2^2 \quad \text{subject to : } \boldsymbol{\lambda} \in \mathbb{R}_{\geq 0}^K.$$

- In IAPF and OAPF, IS weights are *exact*:

$$w_t^{(m)} = \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \sum_{j=1}^K w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^K \lambda_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}$$

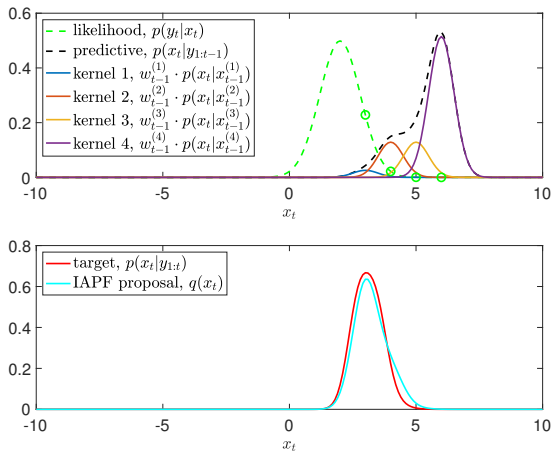
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<sup>20</sup>N. Branchini and V. Elvira. “Optimized auxiliary particle filters”. In: *Uncertainty in Artificial Intelligence*. PMLR. 2021, pp. 1289–1299.

<sup>21</sup>N. Branchini and V. Elvira. “An adaptive mixture view of particle filters”. In: *Foundations of Data Science* (2024), pp. 0–0.



## Toy example: IAPF with $M = 4$ particles



- predictive,  $p(x_t|y_{1:t-1}) = \sum_{j=1}^M w_{t-1}^{(j)} p(x_t|x_{t-1}^{(j)})$  with  $w_{t-1} = [0.03, 0.16, 0.16, 0.65]$
- IAPF proposal,  $\psi_t^{\text{IAPF}}(x_t) = \sum_{j=1}^M \lambda_t^{(j)} p(x_t|x_{t-1}^{(j)})$ , with  $\lambda_t^{\text{IAPF}} = [0.7657, 0.2276, 0.0066, 0.0001]$

## Summary: PF framework from MIS perspective

- (i) Initialization. At time  $t = 0$ ,  $\mathbf{x}_0^{(m)} \sim p(\mathbf{x}_0)$ , and  $w_0^{(m)} = 1/M$ ,  $m = 1, \dots, M$ .
- (ii) Recursive step. At time  $t > 0$ ,
- Proposal adaptation/selection.**<sup>22</sup> Select the MIS proposal of the form

$$\psi_t(\mathbf{x}_t) = \sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)}), \quad \text{with} \quad \lambda_t^{(j)} = ?$$

- Sampling.** Draw samples according to

$$\mathbf{x}_t^{(m)} \sim \psi_t(\mathbf{x}_t), \quad m = 1, \dots, M.$$

- Weighting.** Compute the normalized IS weights by

$$w_t^{(m)} = ?$$

|                   | BPF                                            | APF                                                                                               | IAPF and OAPF                                                                                                                                                                                                  |
|-------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\lambda_t^{(m)}$ | $w_{t-1}^{(m)}$                                | $\propto p(\mathbf{y}_t   \bar{\mathbf{x}}_t^{(m)}) w_{t-1}^{(m)}$                                | $\propto \frac{p(\mathbf{y}_t   \bar{\mathbf{x}}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\bar{\mathbf{x}}_t^{(m)}   \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M p(\bar{\mathbf{x}}_t^{(m)}   \mathbf{x}_{t-1}^{(j)})}$ |
| $w_t^{(m)}$       | $\propto p(\mathbf{y}_t   \mathbf{x}_t^{(m)})$ | $\propto \frac{p(\mathbf{y}_t   \mathbf{x}_t^{(m)})}{p(\mathbf{y}_t   \bar{\mathbf{x}}_t^{(m)})}$ | $\propto \frac{p(\mathbf{y}_t   \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)}   \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t^{(m)}   \mathbf{x}_{t-1}^{(j)})}$   |

- In all PFs:

$$p(\mathbf{x}_t | \mathbf{y}_{1:t}) \approx \sum_{m=1}^M w_t^{(m)} \delta_{\mathbf{x}_t^{(m)}}(\mathbf{x}_t)$$

<sup>22</sup>V. Elvira, L. Martino, M. F. Bugallo, and P. M. Djuric. "Elucidating the auxiliary particle filter via multiple importance sampling [lecture notes]". In: *IEEE Signal Processing Magazine* 36.6 (2019), pp. 145–152.

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- Weighting.** Compute the normalized IS weights by

$$w_t^{(m)} = ?$$

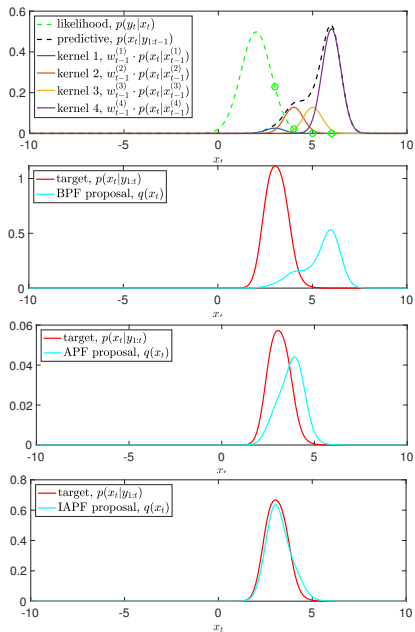
|                   | BPF                                            | APF                                                                                                 | IAPF and OAPF                                                                                                                                                                                                  |
|-------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\lambda_t^{(m)}$ | $w_{t-1}^{(m)}$                                | $\propto p(\mathbf{y}_t   \bar{\mathbf{x}}_t^{(m)}) w_{t-1}^{(m)}$                                  | $\propto \frac{p(\mathbf{y}_t   \bar{\mathbf{x}}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\bar{\mathbf{x}}_t^{(m)}   \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M p(\bar{\mathbf{x}}_t^{(m)}   \mathbf{x}_{t-1}^{(j)})}$ |
| $w_t^{(m)}$       | $\propto p(\mathbf{y}_t   \mathbf{x}_t^{(m)})$ | $\propto \frac{p(\mathbf{y}_t   \mathbf{x}_t^{(m)})}{p(\mathbf{y}_t   \bar{\mathbf{x}}_t^{(i_m)})}$ | $\propto \frac{p(\mathbf{y}_t   \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)}   \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t^{(m)}   \mathbf{x}_{t-1}^{(j)})}$   |

- In all PFs:

$$p(\mathbf{x}_t | \mathbf{y}_{1:t}) \approx \sum_{m=1}^M w_t^{(m)} \delta_{\mathbf{x}_t^{(m)}}(\mathbf{x}_t)$$

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## Toy example: summary



## Numerical result 1: channel estimation in wireless system

- We suppose a linear-Gaussian system described by

$$\mathbf{x}_t = \mathbf{A}\mathbf{x}_{t-1} + \mathbf{r}_t,$$

$$y_t = \mathbf{h}_t^\top \mathbf{x}_t + \mathbf{r}_t,$$

- $\mathbf{h}_t = [h_t, h_{t-1}, \dots, h_{t-d_x+1}]^\top$ , last  $d_x$  transmitted pilots,  $d_t \in \{-1, +1\}$ ,
- $\mathbf{A} = 0.7\mathbf{I}$
- $\mathbf{q}_t \sim \mathcal{N}(0, \mathbf{Q})$ ,  $\mathbf{Q} = 5\mathbf{I}$
- $\mathbf{r}_t \sim \mathcal{N}(0, \mathbf{R})$ ,  $\mathbf{R} = 0.5$
- we set  $T = 200$  time steps and  $M = 100$  particles

| $d_x$ (dimension) | 1             | 2             | 3             | 5             | 10            |
|-------------------|---------------|---------------|---------------|---------------|---------------|
| MSE (BPF)         | 0.0272        | 0.3762        | 0.9657        | 1.4705        | 2.9592        |
| MSE (APF)         | 0.0709        | 0.8041        | 1.6041        | 2.2132        | 3.7187        |
| MSE (IAPF)        | <b>0.0062</b> | <b>0.1764</b> | <b>0.5176</b> | <b>0.8041</b> | <b>2.6931</b> |

# Outline

## State-space models and Bayesian filtering

### Importance sampling: basics and advanced methods

- Monte Carlo basics

- Importance sampling (IS) basics

- Multiple importance sampling (MIS)

- Adaptive importance sampling (AIS)

### Particle filtering

- From IS to sequential IS

- Mini-project

- In search of better particle filters

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## Particle filtering from the MIS and AIS perspectives

### Advanced particle filtering

- PFs in high-dimension spaces

- The choice of  $N$  in PFs

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# Filtering in high-dimension spaces

- All methods, and Monte Carlo is not an exception, suffer from the curse of dimensionality (in this case, when  $d_x$  is high)
- In some models, some of the hidden variables can be integrated (Rao-blackwellized PFs):<sup>23</sup>
  - implicit dimensionality reduction in the latent space
  - lower variance of PF estimators, compared to working in the original space
- Another approach is to partition the space and run several filters in parallel (Multiple PFs):<sup>24</sup>
  - implicit dimensionality reduction in the latent space
  - works well when the dimensions of each subset only interact within each subset
  - more research is needed

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<sup>23</sup>K. Murphy and S. Russell. "Rao-Blackwellised particle filtering for dynamic Bayesian networks". In: *Sequential Monte Carlo methods in practice*. Springer, 2001, pp. 499–515.

<sup>24</sup>P. M. Djuric, T. Lu, and M. F. Bugallo. "Multiple particle filtering". In: *2007 IEEE International Conference on Acoustics, Speech and Signal Processing-ICASSP'07*. Vol. 3. IEEE. 2007, pp. III–1181.



# Outline

## State-space models and Bayesian filtering

### Importance sampling: basics and advanced methods

- Monte Carlo basics

- Importance sampling (IS) basics

- Multiple importance sampling (MIS)

- Adaptive importance sampling (AIS)

### Particle filtering

- From IS to sequential IS

- Mini-project

- In search of better particle filters

- Mini-project

### Particle filtering from the MIS and AIS perspectives

### Advanced particle filtering

- PFs in high-dimension spaces

- The choice of  $N$  in PFs

# Convergence assessment and adapting $N$ in PF

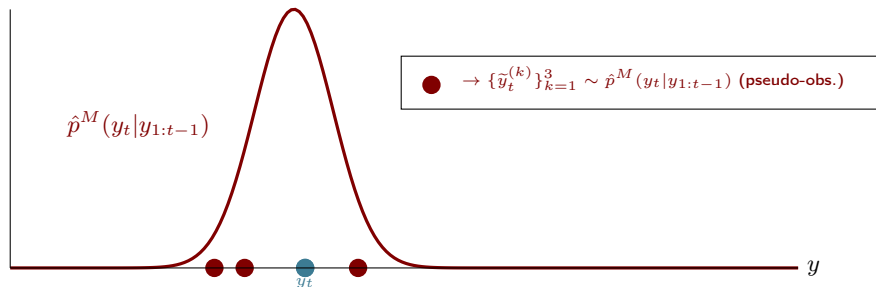
- **Goal:** in real time and for any SSM:
  1. **evaluate the convergence**, related to the quality of the approximation and
  2. **adapt the number of particles**<sup>25</sup>,
  3. with theoretical guarantees<sup>26</sup>
- **Intuition:** check whether the **received observations** “make sense” with the **approximated predictive distributions**
- **Challenge:**
  - at each time step just one observation  $y_t$  available
  - the predictive  $\hat{p}^M(y_t|y_{1:t-1})$  is evolving with time
- **Proposed method:** At each time step  $t$ 
  - Generate  $K$  fictitious observations  $\tilde{y}_t^{(k)}$  from  $\hat{p}^M(y_t|y_{1:t-1})$ 
    1.  $\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t|\tilde{\mathbf{x}}_{t-1}^{(m)})$  (prediction step of BPF, **for free**)
    2.  $\tilde{y}_t^{(k)} \sim \frac{1}{M} \sum_{m=1}^M p(y_t|\mathbf{x}_t^{(m)})$ ,  $k = 1, \dots, K$  (**cheap**  $K \ll M$ )
  - Compare them with the actual observation  $y_t$ .
    - \* Implicitly, we compare  $\hat{p}^M(y_t|y_{1:t-1})$  and  $p(y_t|y_{1:t-1})$

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<sup>25</sup>V. Elvira, J. Míguez, and P. M. Djurić. “Adapting the number of particles in sequential Monte Carlo methods through an online scheme for convergence assessment”. In: *IEEE Transactions on Signal Processing* 65.7 (2016), pp. 1781–1794.

<sup>26</sup>V. Elvira, J. Míguez, and P. M. Djurić. “On the performance of particle filters with adaptive number of particles”. In: *Statistics and Computing* 31 (2021), pp. 1–18.

## Ordering observation and fictitious observations

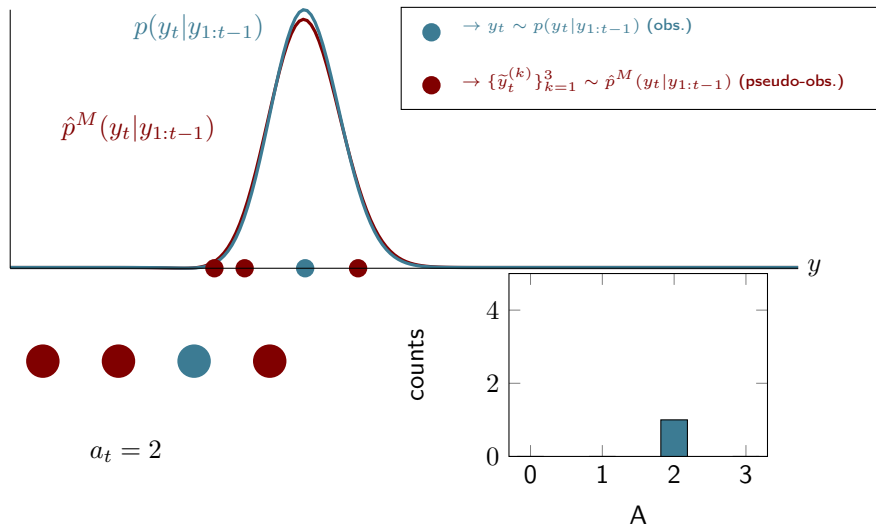


- $A_t$ : number of fictitious observations,  $\{\tilde{y}_t^{(k)}\}_{k=1}^3$ , smaller than  $y_t$

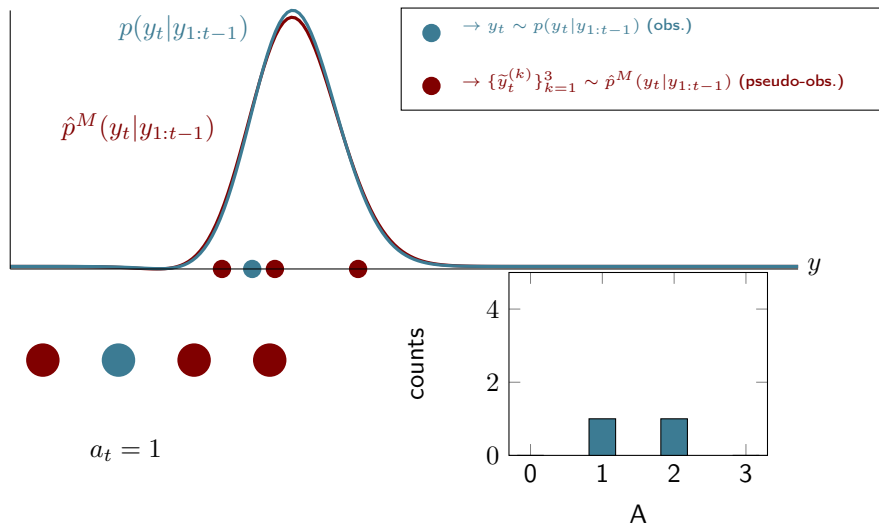


- We can iteratively compute  $a_t$

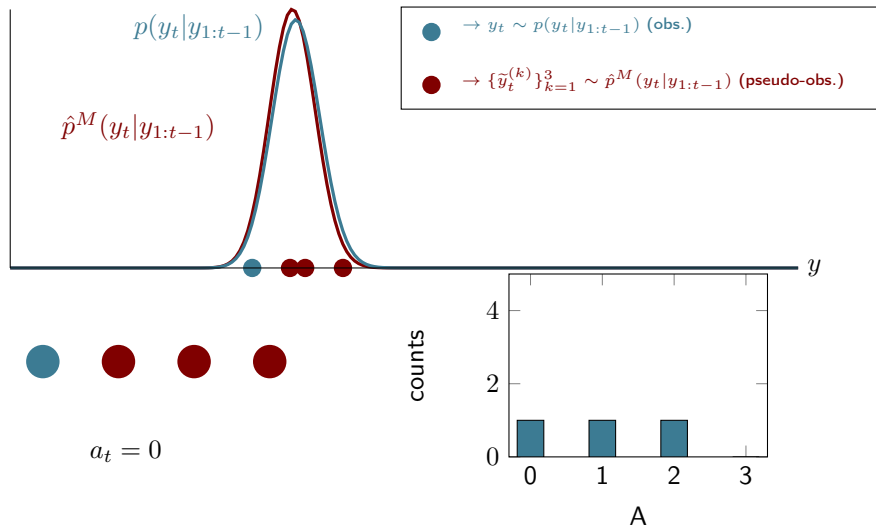
Good approximation,  $t = 1$



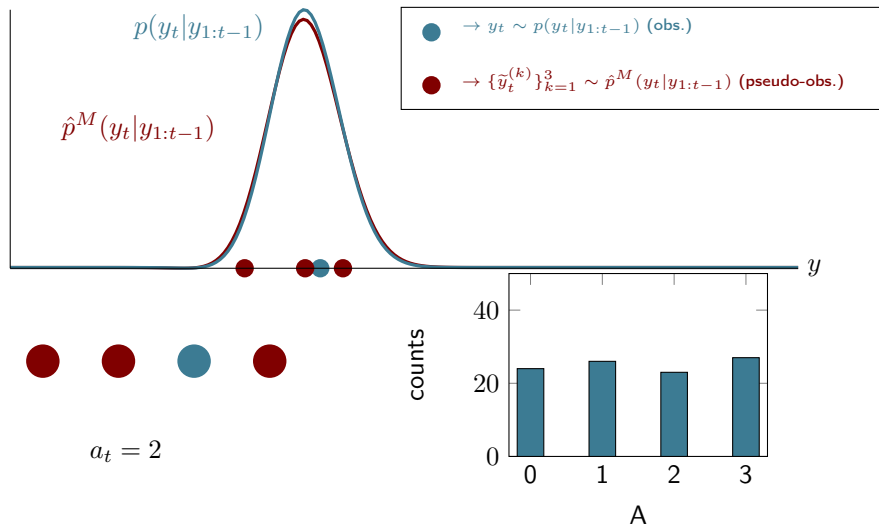
Good approximation,  $t = 2$



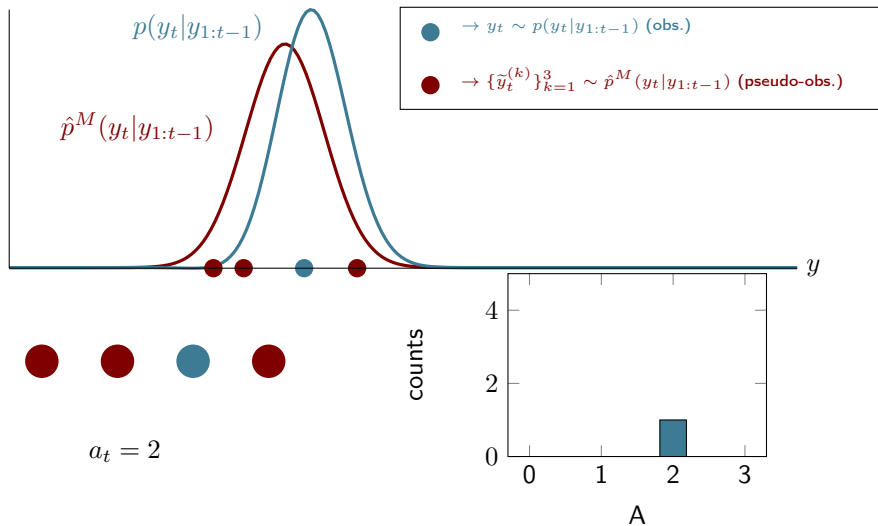
Good approximation,  $t = 3$



Good approximation,  $t = 100$

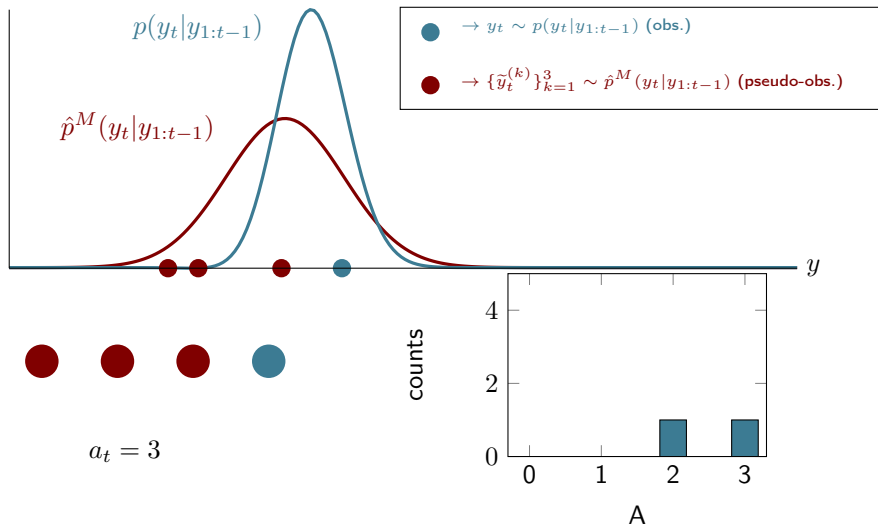


## Bad approximation, $t = 1$

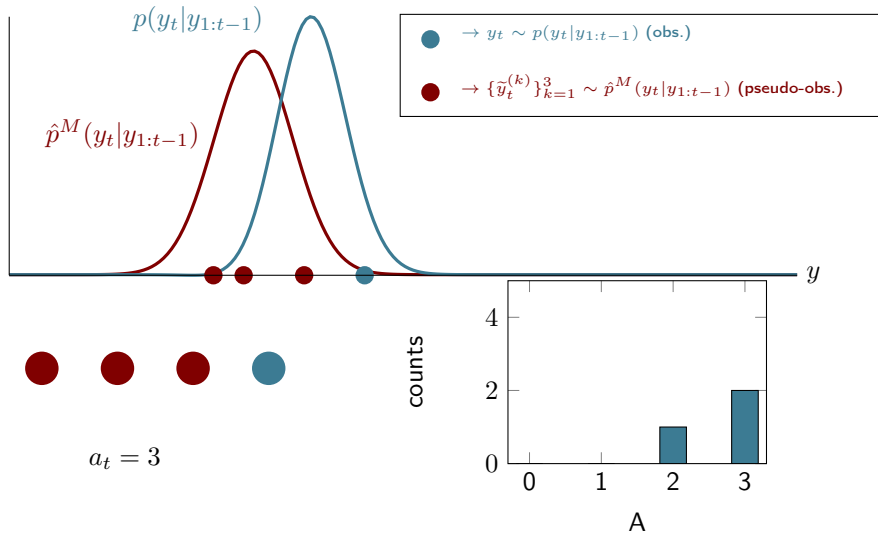




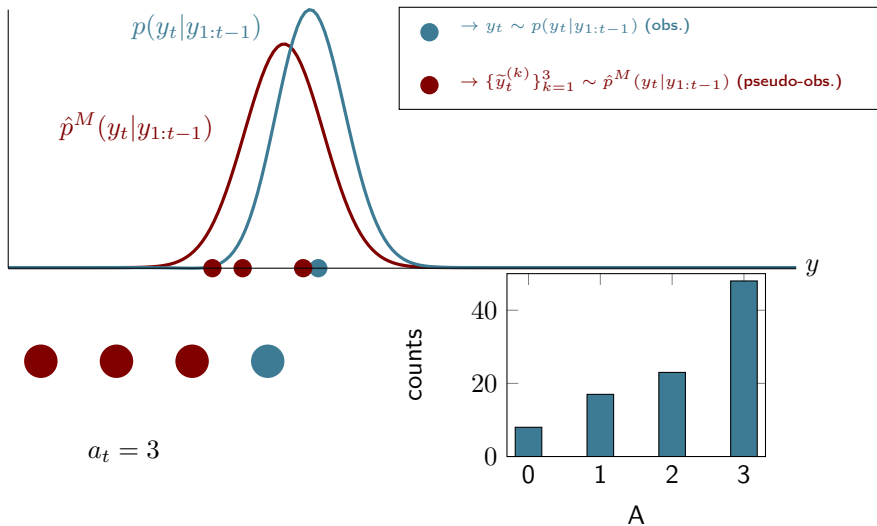
## Bad approximation, $t = 2$



## Bad approximation, $t = 3$



## Bad approximation, $t = 100$



# Methodology summary and properties

- **Methodology:** At each time step:
  - simulate  $\tilde{y}_t^{(k)} \sim \hat{p}^M(y_t | y_{1:t-1})$ ,  $k = 1, \dots, K$
  - build the r.v.  $A_{K,t} := |\mathcal{A}_{K,t}| \in \{0, 1, \dots, K\}$ , where  $\mathcal{A}_{K,t} := \{y \in \{\tilde{y}_t^{(k)}\}_{k=1}^K : y < y_t\}$
- **Properties:** Under the hypothesis of **perfect approximation**:
  - $\mathcal{J}_t := \{y_t, \tilde{y}_t^{(1)}, \dots, \tilde{y}_t^{(K)}\}$  is a set of i.i.d. samples from a **common** continuous probability distribution  $p_t(y_t)$ , then:

**Proposition 1:** *the pmf of the r.v.  $A_{K,t}$  is **uniform**:*

$$\mathbb{Q}_K(n) = \frac{1}{K+1}, \quad n = 0, \dots, K.$$

**Proposition 2:** *the r.v.'s  $A_{K,t_1}$  and  $A_{K,t_2}$  are **independent**,  $\forall t_1, t_2 \in \mathbb{N}$  with  $t_1 \neq t_2$ .*

- **Invariant wrt the state space model!**

# Theoretical results

- **Theoretical analysis:**

- **convergence** of the predictive pdf of the observations:<sup>27</sup>

$$\lim_{M \rightarrow \infty} \left( f, \hat{p}^M(y_t | y_{1:t-1}) \right) = \left( f, p(y_t | y_{1:t-1}) \right) \quad \text{a.s.,}$$

with explicit **convergence rate**

- ★ extends the existing results of pointwise convergence of  $\hat{p}^M(y_t | y_{1:t-1})$  to  $\hat{p}(y_t | y_{1:t-1})$
- ★ holds for **multidimensional** observations
- ★ key for the statistical analysis of  $A_{K,t}$
- **convergence** of the p.m.f. of  $A_{K,t}$  to a discrete uniform distribution

$$\frac{1}{K+1} - \varepsilon_M \leq \mathbb{Q}_K(n) \leq \frac{1}{K+1} + \varepsilon_M, \quad n = 0, \dots, K,$$

with  $\lim_{M \rightarrow \infty} \varepsilon_M = 0$  a.s.

- Uniformity of the statistic  $A_{K,t}$  (with  $K$  fictitious observations) equivalent to  $\hat{p}^M(y_t | y_{1:t-1})$  and  $\hat{p}(y_t | y_{1:t-1})$  matching  $K$  moments.<sup>28</sup>

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<sup>27</sup>V. Elvira, J. Míguez, and P. M. Djurić. “Adapting the number of particles in sequential Monte Carlo methods through an online scheme for convergence assessment”. In: *IEEE Transactions on Signal Processing* 65.7 (2016), pp. 1781–1794.

<sup>28</sup>V. Elvira, J. Míguez, and P. M. Djurić. “On the performance of particle filters with adaptive number of particles”. In: *Statistics and Computing* 31 (2021), pp. 1–18. 